

AC BRUSHLESS

(Synchronous)

FIRST studied "synchronous"

AC BRUSHLESS [synchronous machine]

(MACHINE)

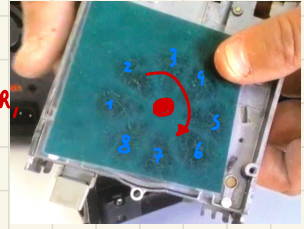


ROTOR with permanent magnet sectors

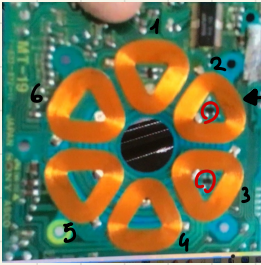
FLOPPY DISK:

(different solutions...)
(used for example for floppy disk)

moving the ROTOR,
8 poles, ($m_p = 4$)
of the machine
(pole movement when Rotor moves)



COILS of the three phase machine

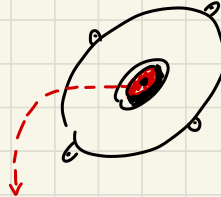


(6 windings)
(strange approach to design coils)

→ inside the coils there is an hall effect sensor, to produce the controller
(so to recognize N/S)



PLATE to close the machine ROTOR
↓
(close the flux) inside the machine



STATOR



permanent magnet around the external cylinder

ROTOR



→ typical of CD ROM computer machine...
(condensed indromes also)

+ to supply LED + temperature sensor to check if hot (by looking LED light)

2 main families:

(BRUSHLESS MACHINE)

{ AC · Focus of that Lesson...
DC

In short circuit there is a torque $\neq$ respect induction machine

$\downarrow$
there is some permanent magnet, producing electromotive force even in short circuit! $\leftarrow$ some element producing emf with movement

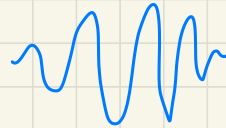
EXPERIMENTS

- ① NOT supplied: move ROTOR and measure STATOR terminals:
try to measure the **Voltage** between two terminals.. when moving ROTOR



(Rotate the ROTOR..)

you see some effect

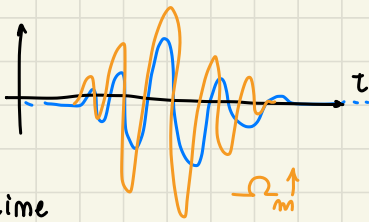


(electromotive force..)

IF ROTATING the ROTOR faster by hands $\Omega_m \uparrow$ it has effect:

$\left\{ \begin{array}{l} \text{increase freq of} \\ \text{electrical quantity} \\ \text{and amplitude of} \\ \text{electromotive force} \end{array} \right\}$

$\Omega \uparrow \Rightarrow$ module of emf increases and also freq increases $\rightarrow$ typical of synchronous machine



In which $\omega \sim \Omega_m$ and $e_{ms} \sim \Omega_m$

- ② short circuit the stator windings

short circuit current:

$\downarrow$ move the ROTOR and measure current on windings

[properly connecting the machine and probes..]

When moving the ROTOR

high current in short circuit condition:

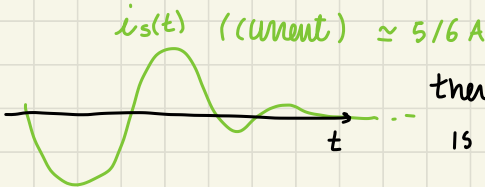
there is an emf and on stator coils there

is an impedance $\rightarrow$ small Reactance...

coils Resistance and inductance

$\downarrow$ so in short circuit ATTENTION...

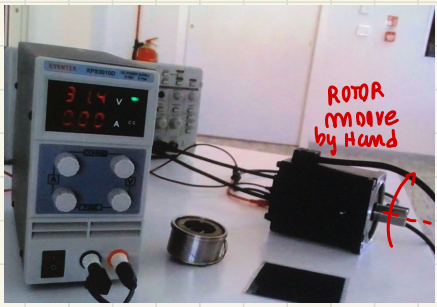
i_{CORTO} VERY HIGH!
CIRCUITO



$\downarrow$
High current **Dangerous!** may destroy the machine

3) Supply STATOR coils

• another TEST: apply the voltage by power supply



supply 31 V $\rightarrow$ between two terminals of STATOR
 $\downarrow$
 [there is an inductor $\rightarrow$ because I can see some "sparks" when connected due to di/dt]
 reduce VOLTAGE supply:

stable positions of the ROTOR when remain fixed $\rightarrow$ IF I try to move the ROTOR.

in 5 positions aligned magnet (NORTH of ψ stator $\leftarrow$ current aligned with SOUTH of ROTOR) $\rightarrow$ m_p used to see that 1 period mechanical is 5 electrical period (to realize 1 turn, electrical quantity perform 5 turns)

there are 5 STABLE POSITIONS (where ROTOR is $\rightarrow (m_p = 5 !)$ hard to rotate) (mechanical turn)

(move from mech world speed to electrical through m_p)

4) by changing the phase supplied in STATOR, moving the coils:

if supply properly the coil by (120°) electrically
 (so mechanically $120/5 \approx 24^\circ$)

$\searrow$ changing supplied coils the ROTOR moves!

$\downarrow$ on this machine there is some shaft oscillation... (NOT smoothed machine, oscillating)
 $\downarrow$ Great DIFFERENCE

$\downarrow$ move electrical quantity by 120° correspond to $120/m_p = 24^\circ$ on mechanical world

(synchronous) $\neq$ induction (short circuit ROTOR: very stable ψ_r , smooth)

$\downarrow$ different approach fast response, no smoothing $\Rightarrow$ I'm able to move rotor, position defined by stator flux position (move ROTOR changing supply phase) (ROTOR align, moving)

you can have vibration without torque if loose the step

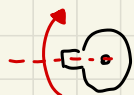
$\leftarrow$ { with suitable P.S I can properly } drive the motion

CAREFULL! if start immediately with high speed ψ_s , ROTOR is unable to lock to stator $\rightarrow$ it lose the STEP ("perdita di passo" della macchina sincrona)

drive well the magnetic flux given by stator currents and ROTOR move...

- starting with high stator flux speed, ROTOR unable to follow: "pendule de passé"
you have vibrations on ROTOR without any torque ! (BAD)
- When ROTATING ROTOR from stable position

initially low torque... then $T_e \uparrow$ and then $T_e < 0$ go
on next position



behaviour typical of e.m. Joint

torque becomes $T_e < 0 \rightarrow$

generic
propert. $=: K$

T_e proportional to the angle δ

$$\left\{ T_e = \frac{m_p}{L_{ks}} |\psi_s| |\psi_r| \sin(\delta) \right\}$$

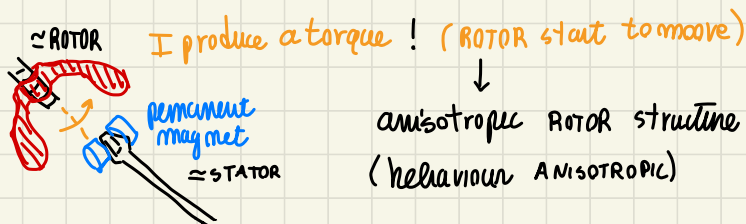
GIVEN by PERMANENT magnet $\psi_r = \psi_m$

(typical of
AC brushless)

proportional to angle
between ROTOR/STATOR flux
and permanent magnet ψ_r flux

- and as you can see, IF you increase current: $|\psi_s| \uparrow \Rightarrow T_e \uparrow$ you are not
able to move the ROTOR by hand! and if $i_s \downarrow \Rightarrow T_e \downarrow$

$\Rightarrow$ similar behaviour on this SYSTEM



anisotropic ROTOR structure
(behaviour ANISOTROPIC)

ON OUR EXPERIMENT

IM AC Brushless: permanent magnet Rotor (because in short circuit we
perceive torque)

BUT we need to test properly to check for ROTOR dissymmetry and see if
it is ISO/anisotropic $\rightarrow$ you need particular test!



$\hookrightarrow$ ANISOTROPIC PM AC a lot used...

ISOTROPIC / ANISOTROPIC

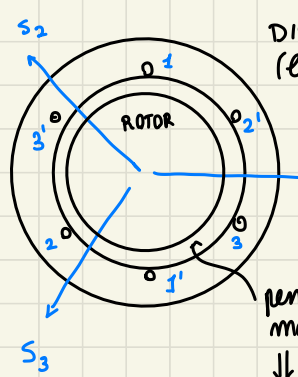
With permanent magnet AC Brushless machine

AC BRUSHLESS (synchronous machine with permanent magnet on the rotor)

[03/05]

only 3 terminals, we see as an equivalent circuit Δ, Y inside $\rightarrow$ (while 6 on induction machine)

We don't know if inside it's connected in Δ, Y , we think on model



Distributed coils on stator (like induction machine) $\rightarrow$ [STATOR in AC brushless (equals to induct. mach)]

We highlight (magnetic axis of the coils)

permanent magnet rotor (without rotor coils $\neq$ from induction machine) [ROTOR in AC Brushless...]
 $\downarrow$ different solutions:

Different solutions (to realize Rotor)

[4 poles example of IPM]

(not used)

permanent magnet fixed on the external surface of the rotor

each of this:

SM PM

[surface mounted permanent magnet]

(ISOTROPIC)

(hard to build, material of PM easy break)

fe material sheet

[permanent magnet inside the rotor]

IPM

[Interior permanent magnet]

{ better mechanically, it is fixed inside! more robust }

(ANISOTROPIC)

punching technique to fix PM inside hole

(ANISOTROPIC)

(4 poles example)

Air

Synchronous Reluctance Machine

[synch Rel]

(NOT a lot used!)

(4 poles example)

permanent magnet inside the Air gap

Air

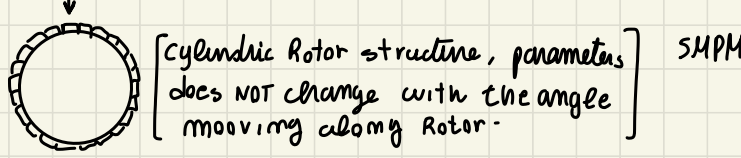
[APM synch Rel]

Synchronous Reluctance Machine Assisted by Permanent Magnet

(lot used in electric cars motor)

(ANISOTROPIC)

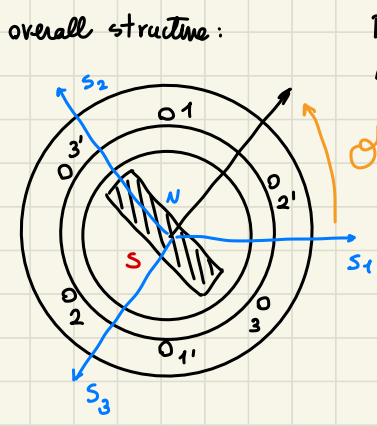
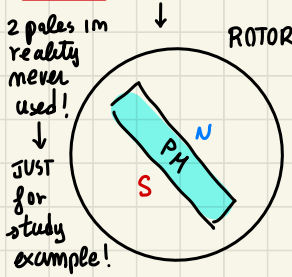
on SMPM: that machine has a symmetrical rotor structure, without strange effect as function of mechanical angle: "ISOTROPIC" structure



While the others have strange behaviour with angle → "ANISOTROPIC" structure

[SymchRel] → complex theory
 [IPM] / (APM symchRel) → similar modelling POV
 ↓ initially we focus on SMPM / IPM / SRHAPM study

IPM (2 poles machine example) [ANISOTROPIC machine] (more generic structure)



During simple testing we are not able to identify the machine structure SMPM/IPM?

↙ We consider this model ... of the machine!

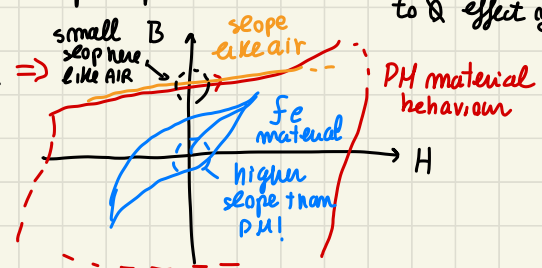
Permanent Magnet (PM) (Behaviour)

↓ generates a Flux.. "generator of magnetic Flux!"

In the P.O.V of the other currents... IF you need to apply superimposition principle considering singularly each coil current (of stator current)

by superimposition principle ⇒ Considering only effect of 1st coil you need to reduce to effect of other coils flux and also effect of PM (all other causes to Q!)

• IF PM effect to Q ⇒ (neglect PM behaviour, only consider it like air)



you have to consider it like AIR (not like Fe)

So when considering superimposition principle... you need to consider instead of PM a block of AIR (when switching OFF its effect!)



When consider only i_{s1} Effect $\rightarrow$ PM as AIR

flux lines... to compute self inductance.

AIR = large reluctance $\rightarrow$ flux lines move on the boundaries, avoid the AIR
AIR has larger Reluctance, flux don't go inside through the air

SO IF ROTOR MOVES, the PM (AIR now) move:

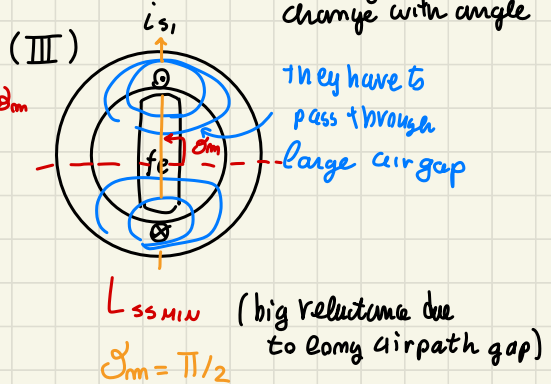
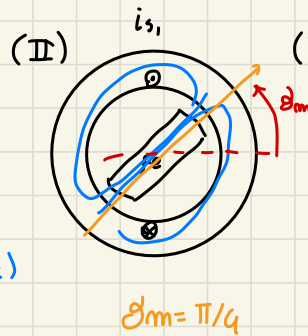
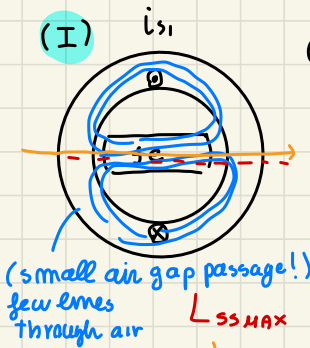
so it means that self inductance depends on Rotor position!

To understand it:

IF we show Rotor as (strange Rotor) different structure
IM different ROTOR POSITION:

[synchronous machine
with bounded Rotor]

anisotropic structure
That show us better
how self inductance
change with angle



Where the self inductance is maximum? (I)

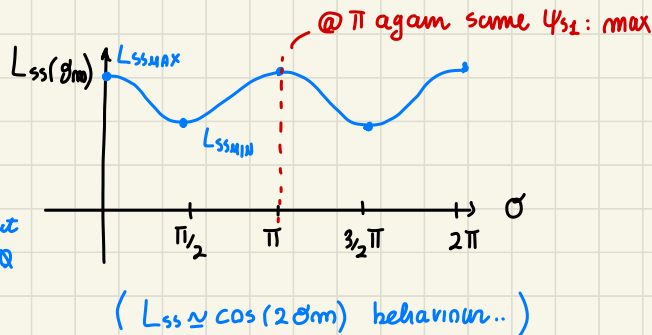
being it proportional to the number of turns squared $\times$ permeance
where:
(small air gap means low reluctance $\rightarrow$ high inductance)
(divided by reluctance)

SELF INDUCTANCE

flux linked with first coil

$$L_{ss} = \frac{\Psi_{s1}}{i_{s1}}$$

when all this effect set to 0
 $i_{s2} = 0$
 $i_{s3} = 0$
 $\Psi_{PM} = 0$



mutual inductance

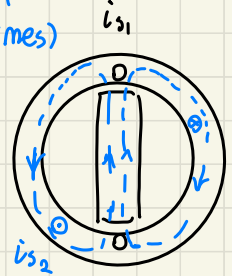
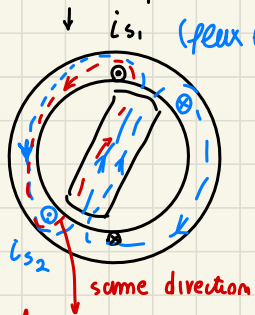
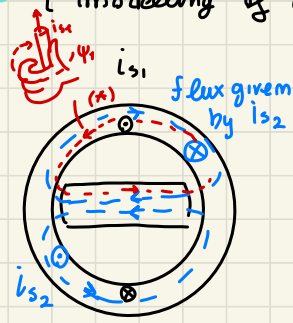
[there are some problems on the modelling of this machine (middle position)]

move where lower air gap

$$M_{ss} = \frac{\Psi_{s1}}{i_{s2}}$$

$i_{s1} = 0$
 $i_{s3} = 0$
 $\Psi_{pH} = 0$

we look to the flux given by i_{s2} linked with s_1 coil



$M_{ss} \geq 0$?

$M_{ss} > 0$ same flux direction respect i_{s1} and i_{s2}

{ there is M_{ss} oscillation! }

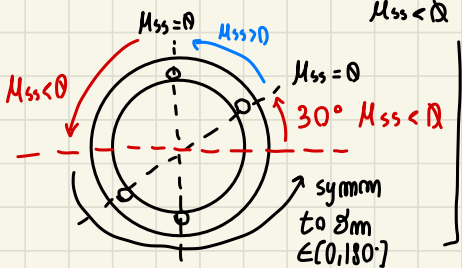
$\Psi_{s1} = 0$ (NO linkage)

flux given by i_{s2} on coil s_2 when $\theta_m = \pi/2$ is 0, because no flux lines which include i_{s2} coil

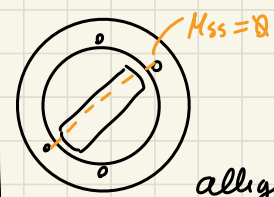
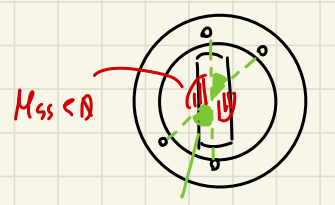
its sign depends: IF flux given by i_{s2} has same direction of flux given by i_{s1} means that $M_{ss} > 0$

here flux lines move on opposite direction! $M_{ss} < 0$

(i_{s1} has opposite effect to i_{s2}) for $\theta_m = 0$ $M_{ss} < 0$

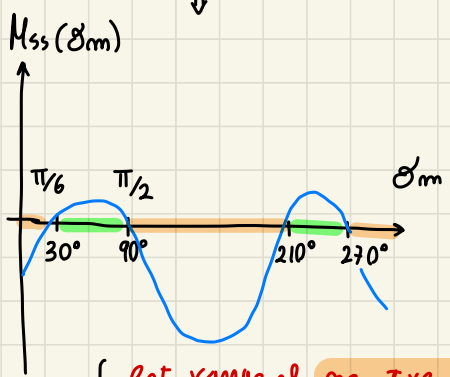


$M_{ss} > 0$ When Rotor im between 2 position



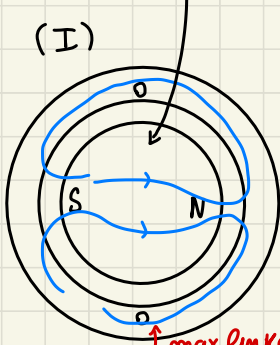
aligned with s_2 there is no linkage between

because here < 0 here > 0 in the middle $M_{ss} = 0$

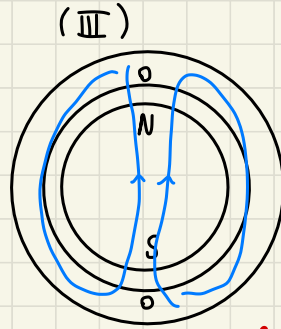
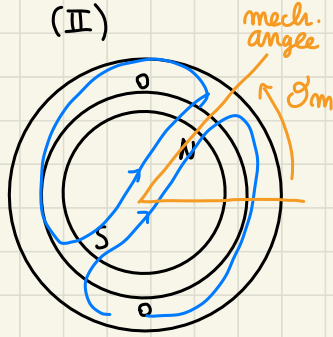


{ not range of negative M_{ss} in some range positive }

considering Ψ_{PM} (permanent magnet inside the ROTOR producing lines)
(don't look to the ROTOR structure...) (EFFECT of PM)



$\Psi_{PM\max}$ max linkage with $s_1 \rightarrow$ lot linked lines



$\Psi_{PM} = 0 \rightarrow$ no lines linked with s_1

permanent magnet effect
independently from ROTOR PM structure it produces this kind of flux lines as 2 poles...

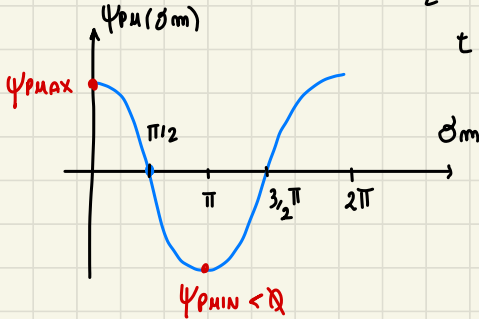
Ψ_{PM} linked with s_1 ?

- (I) max linkage : $\Psi_{PM\max}$
- (III) $\Psi_{PM} = 0$ no lines linked

When ROTOR move by $\theta_m = \pi$

lot linked lines but now neg linkage between...

$$\Psi_{PM\min} = -\Psi_{PM\max}$$



so overall we obtain that Ψ_{PM} flux..

cosinusoidal waveform thanks to distributed windings!

$$\Psi_{PM} = \hat{\Psi}_{PM} \cos(\theta_m)$$

true only for AC brushless

[due to machine design with distributed windings]

in AC brushless, the distributed winding guarantee Ψ_{PM} cosinusoidal

(Now that we find the inductances)

$\rightarrow$ we can introduce machine dyn equations... $\Rightarrow$

Dyn equations of AC Brushless (IPM)

$$V_{s1} = R_s i_{s1} + p \psi_{s1}$$

$$V_{s2} = R_s i_{s2} + p \psi_{s2}$$

$$V_{s3} = R_s i_{s3} + p \psi_{s3}$$

same stator coil equations as induction machine...
but different ψ_s equation...

coil equation!

(NOT 6 terms as IM)

difference w.r.t induction machine

now ψ_s as function of mech angle θ_m !

max when $\theta_m = +\frac{2}{3}\pi$

now only 4 elements as function of θ_m (ANISOTROPIC)

$$\psi_{s1} = L_{ss}(\theta_m) i_{s1} + M_{ss}(\theta_m) i_{s2} + M_{ss}(\theta_m - \frac{2}{3}\pi) i_{s3} + \psi_{PM}(\theta_m)$$

(*) $\downarrow$ simplifications $\longrightarrow$

$$\psi_{s1} = L_s i_{s1} + \psi_{PM}(\theta_m)$$

$\hookrightarrow L_s = L_{ss} - M_{ss} :=$ Synchronous Inductance

$$\psi_{s2} = L_s i_{s2} + \psi_{PM}(\theta_m - \frac{2}{3}\pi)$$

When $\theta_m = \frac{2}{3}\pi$
you have max effect of PM on $s2$

$$\psi_{s3} = L_s i_{s3} + \psi_{PM}(\theta_m + \frac{2}{3}\pi)$$

$\theta_m = -\frac{2}{3}\pi$
guarantee max PM effect on $s3$

$\uparrow$ machine dyn model!

$\downarrow$ in this form seems of III order but because $i_{s1} + i_{s2} + i_{s3} = 0$

in reality it is II ORD $\rightarrow$ [2 var. independent] and ψ_{PM} is function of θ_m ...

$\downarrow$

We wanna get a better viewpoint...

Apply Space Phasor Formula (Clark)

$$\bar{V}_s^{\alpha\beta} = R_s \bar{I}_s^{\alpha\beta} + p \bar{\psi}_s^{\alpha\beta}$$

$\hookrightarrow$ equivalent machine in 2 coils! $\Rightarrow$

We need to introduce simplifications...

1) once introduced anisotropy, we consider now isotropic $\rightarrow$ [and then re-introduce anisotropy later]

$\downarrow$
(ISOTROPIC Machine)

$\left\{ \begin{array}{l} L_{ss} \text{ is constant} \\ M_{ss} \text{ is constant} \end{array} \right. !$

only ψ_{PM} function of θ_m

$$\psi_{s1} = L_{ss} i_{s1} + M_{ss} i_{s2} + M_{ss} i_{s3} + \psi_{PM}(\theta_m)$$

2) Connection of the coils:

• star connection with disconnected star center $\uparrow$ star point
OR • Delta connection $\downarrow$

in both cases: $i_{s1} + i_{s2} + i_{s3} = 0$

[due to the connection]

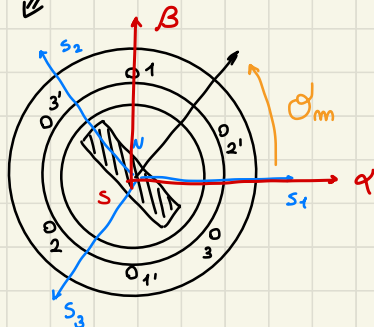
collecting $M_{ss} \downarrow [i_{s2} + i_{s3} = -i_{s1}]$

$$\psi_{s1} = (L_{ss} - M_{ss}) i_{s1} + \psi_{PM}(\theta_m) \text{ we use this form! (*)}$$

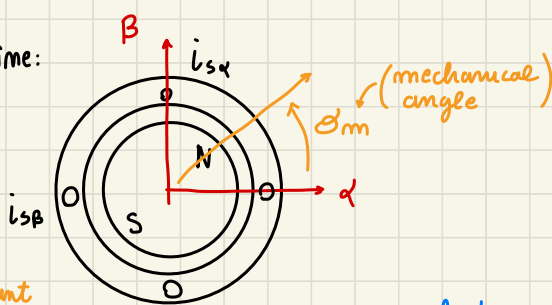
to look at all quantities inside the machine,

the Best viewpoint?

(STATOR? ROTOR? which is Best P.O.V.)



by means of that formula we introduce new fictitious 2 coil machine:
(same as induction machine...)



$$\bar{V}_s^{d\beta} = R_s \bar{I}_s^{d\beta} + p \bar{\Psi}_s^{d\beta}$$

$$\bar{\Psi}_s^{d\beta} = L_s \bar{I}_s^{d\beta} + \sqrt{\frac{3}{2}} \hat{\Psi}_{PM} e^{j\theta_m} \quad \text{constant} \quad \text{coefficient}$$

space phasor formula to cosinusoidal waveform with 120° displacement

when applied on voltages: OVERALL you obtain (amplitude is the RMS value of line to line V) [for $\theta_m = 0 \rightarrow$ max linked flux $(\sqrt{3/2} \hat{\Psi}_{PM})$ by the coil!]
 $e^{j\theta_m} = 1$ $1/\sqrt{2}$ RMS, $\sqrt{3}$ by line to line

↑ Reg frame fixed with stator view point... permanent magnet flux is a function of mech angle θ_m (sin along d, cosine along β)
(no dependency on MECH angle)

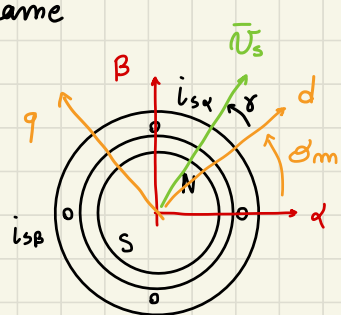
↓ another Ref frame to avoid θ_m dependency $\rightarrow$ looking Respect Rotor
you see Ψ_s fixed not depend on θ_m (constant flux in ROTOR P.O.V)
(ROTATE reference frame on (d,q) axis)

$\Rightarrow$ Rotating by $\theta_s = \theta_m$ (d,q) axis ref frame

d aligned with magnetic North (of PM)

$$\bar{V}_s^{d\beta} = \bar{V}_s^{dq} e^{j\theta_m} \quad \text{to go from dq to } \alpha\beta, \text{ multiply by } e^{j\theta_m} \quad (\theta + \theta_m) \text{ is the angle in } \alpha\beta$$

(to go between the two Ref frame) adding θ_m



↓ so rewrite model in dq Axis:

$$\bar{V}_s^{dq} e^{j\theta_m} = R_s \bar{I}_s^{dq} e^{j\theta_m} + p(\bar{\Psi}_s^{dq} e^{j\theta_m}) = R_s \bar{I}_s^{dq} e^{j\theta_m} + \underbrace{J \dot{\theta}_m \bar{\Psi}_s^{dq} e^{j\theta_m}}_{\text{motional term}} + e^{j\theta_m} p \bar{\Psi}_s^{dq}$$

$$\bar{V}_s^{dq} = R_s \bar{I}_s^{dq} + \underbrace{J \dot{\theta}_m \bar{\Psi}_s^{dq}}_{\text{motional term}} + p \bar{\Psi}_s^{dq}$$

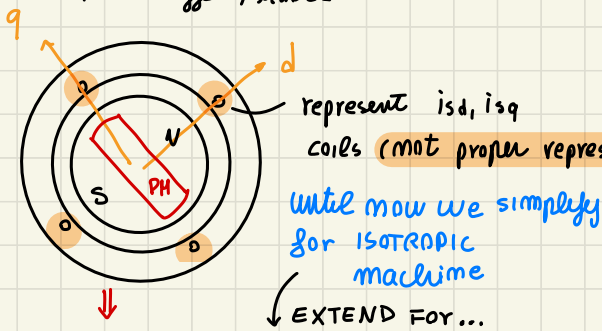
(+ $J \dot{\theta}_m$ motional term as we found) on Induction machine

$$\bar{\Psi}_s^{dq} e^{j\theta_m} = L_s \bar{I}_s^{dq} e^{j\theta_m} + \Psi_{PM} e^{j\theta_m}$$

and overall only 2 currents i_{sd}, i_{sq} (no more 3)

$\Rightarrow$ (dq) OVERALL model in dq REF $\left\{ \begin{array}{l} \bar{V}_s = R_s \bar{I}_s + J \dot{\theta}_m \bar{\Psi}_s + p \bar{\Psi}_s \\ \bar{\Psi}_s = L_s \bar{I}_s + (\hat{\Psi}_{PM}) \end{array} \right.$ (Good view point, where parameters does NOT depend on θ_m !)
stay on Real axis (taken as d)

Final effect, MODEL:

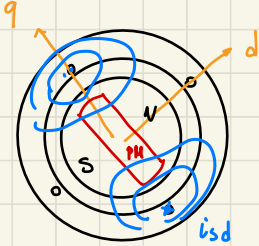


(like armature coil in reality)
they are distributed windings we represent it just as a modelling
(not correct!)
(simplification on what happens inside the machine)

ANISOTROPIC (structure effect)

in this case, i_{sd} self inductance

has lines that when moving rotor they doesn't changes... self inductance of d axis is constant (from that P.O.V d-q) with that (d-q) axis... same for q axis
 $\Rightarrow L_d, L_q$ constant!



they are constant, But:

(for ISOTROPIC $L_d = L_q = L_s$) / ANISOTROPIC machine $L_d \neq L_q$

$L_d \geq L_q$?

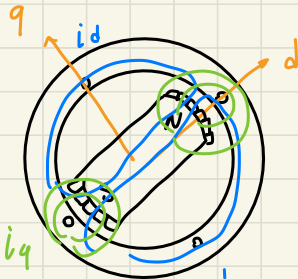
on that configuration...



flux given by i_{sq} go through small air gap...
lot lines pass through small air gap!

While i_{sd} go through longer air gap, due to PH like air (superimposition)

$L_d < L_q$ (on this type of machine!)



• In this case... for a surface mounted magnet rotor (still anisotropic)

L_q pass through larger air gap!

$\Rightarrow$ so... $L_d > L_q$

$\left\{ \begin{array}{l} L_q \text{ goes through } \text{LARGE air gap} \\ \text{respect } L_d \end{array} \right.$

$L_d \neq L_q$ but the relationship $\geq$ depends on ROTOR DESIGN



In a classical
ROTOR structure (4 poles)
(IPM)

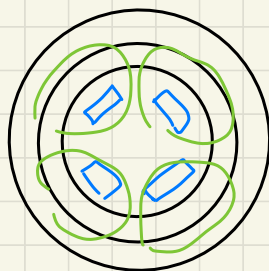
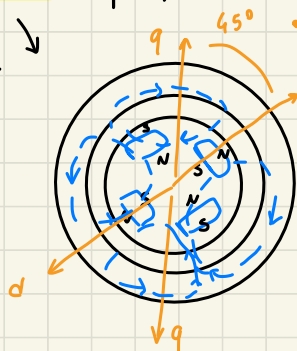
on electrical P.O.V d and q are orthogonal

45° • still $d \perp q$ 90° electrically, but mechanically...
(from mech P.O.V $\downarrow$ d, q with $\approx 45^\circ$ angle
 d, q has $1/2$ the elect angle)

d axis flux lines (stronger lines)

Lines of $d \rightarrow$ large air gap given by PM which behaves like air

[4 POLE
IPM]



$L_d < L_q$

Lines of q axis
small
air gap

$\left\{ \begin{array}{l} \text{Classical} \\ \text{IPM} \\ \text{machine} \end{array} \right.$

(same on APM synchRel)

to show the effect of this anisotropy on Dym. equations...

ANISOTROPIC machine (dym model) $\rightarrow$ same model! no difference on modelly POV

$$\begin{cases} \bar{V}_s = R_s \bar{I}_s + p \bar{\Psi}_s + J \dot{\theta}_m \bar{\Psi}_s \\ \Psi_{sd} = L_d i_{sd} + \Psi_{PM} \\ \Psi_{sq} = L_q i_{sq} \end{cases}$$

(permanent magnet flux linked only on d axis)
 $\hookrightarrow$ no effect on q axis (we see on Ref frame of ROTOR $d \equiv Re$)

$\uparrow$ this model can be used also for ISOTROPIC (ISOTROPIC: $L_d = L_q = L_s$) same dym eq. with this property! while here $L_d \neq L_q$ we should divide it...

TORQUE: to find it we apply **POWER/ENERGY Balance** expression (as done for induction machine) $\Downarrow$

total power going inside machine (similar to induction machine) (here with STATOR motional term)

$$\bar{V}_{sd} i_{sd} + \bar{V}_{sq} i_{sq} = \text{Re} \{ \bar{V}_s \bar{I}_s \}$$

$\downarrow$ using model...

$$\text{Re} \{ \bar{V}_s \bar{I}_s \} = \underbrace{\text{Re} \{ R_s \bar{I}_s \bar{I}_s \}}_{R_s I_s^2 \text{ JOULE LOSSES}} + \underbrace{\text{Re} \{ p \bar{\Psi}_s \bar{I}_s \}}_{i_{sd} p \Psi_{sd} + i_{sq} p \Psi_{sq} =} + \underbrace{\text{Re} \{ J \dot{\theta}_m \bar{\Psi}_s \bar{I}_s \}}_{P_m \text{ (mechanical power)}}$$

Energy stored in inductances

$$\begin{cases} U_{sd} = \frac{1}{2} L_d i_{sd}^2 \\ U_{sq} = \frac{1}{2} L_q i_{sq}^2 \end{cases} \Rightarrow \frac{dU_{sd}}{dt} + \frac{dU_{sq}}{dt}$$

(no effect of energy stored on PM derivative, const)

derivative of total energy stored into inductances

$$P_m = \text{Re} \{ J \dot{\theta}_m \bar{\Psi}_s \bar{I}_s \} = - \dot{\theta}_m \text{Im} \{ \bar{\Psi}_s \bar{I}_s \} = \dot{\theta}_m \text{Im} \{ \bar{I}_s \bar{\Psi}_s \}$$

$$T_e = \frac{P_m}{\Omega_m} = m_p \text{Im} \{ \bar{I}_s \bar{\Psi}_s \}$$

$$\uparrow \dot{\theta}_m = m_p \Omega_m$$

$\downarrow$ BUT continue computation... taking Im per... substitute Ψ_{sd}, Ψ_{sq} from model

$$T_e = m_p (i_{sq} \Psi_{sd} - i_{sd} \Psi_{sq}) \Rightarrow$$

[it is equal to the DTC estimation of the torque! (based on this eq.)]

$$\Rightarrow T_e = m_p \left[i_{sq} (L_d i_{sd} + \psi_{PM}) - i_{sd} L_q i_{sq} \right] = m_p \left[\underbrace{(L_d - L_q) i_{sd} i_{sq}}_{\text{RELUCTANCE TORQUE}} + \underbrace{\psi_{PM} i_{sq}}_{\text{EXCITATION TORQUE}} \right]$$

(collect properly) ↑

disappear when I move the screw and PM, such that ISOTROPIC structure

I'm an
ISOTROPIC
machine

AC BRUSHLESS

ISOTROPIC

$$T_e = m_p (\psi_{PM} i_{sq}) \quad (L_d = L_q = L_s \Rightarrow \text{Reluctance torque goes to } 0)$$

(ψ_{PM} given by permanent magnet)

↑ similar expression!

I'm

RFOC:

$$T_e = m_p (\psi_r i_{sq})$$

INDUCTION

ROTOR FLUX

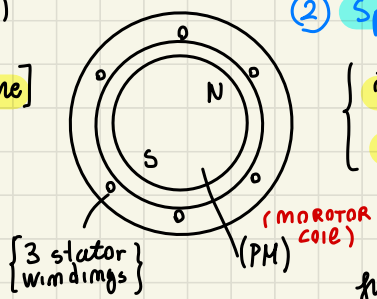
MACHINE

(complex machine) with lots of coils on equation
(given by rotor currents on Rotor 3phase windings)

(on T_e expression.. AC Brushless ISOTROPIC $\sim$ Induction machine) similar depends on Rotor flux obtained by different sources

→ next we have to study control scheme to control torque and flux on AC Brushless... $\Rightarrow$

(we start from)
SMPM
[ISOTROPIC STRUCTURE]



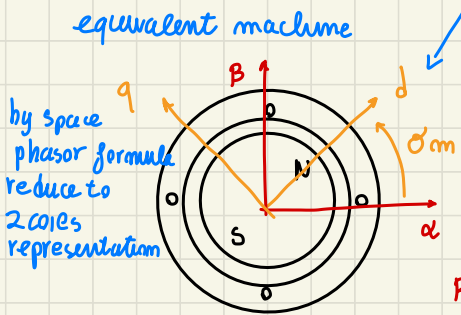
② Space phasor formula

$$\begin{cases} \bar{V}_s = R_s \bar{I}_s + p \bar{\Psi}_s \\ \bar{\Psi}_s = L_s \bar{I}_s + \Psi_{PM} e^{j\theta_m} \end{cases} \quad (\alpha \beta \text{ REF.})$$

$\Psi_{PM} = \sqrt{\frac{3}{2}} \hat{\Psi}_{PM}$

from the dyn eq. initially

①
$$\begin{cases} V_{s1} = R_s i_{s1} + p \Psi_{s1} \\ \dots \\ \Psi_{s1} = \dots \text{ (4 parameters } i_{s1,2,3}, \Psi_{PM} \end{cases}$$
 lot of parameter!



reduces the structure to a 2 windings equivalent machine → (ORTHOGONAL $\alpha \beta$ coils) (NO mutual inductance)

③ Introduce reference ROTATION
ROTATION $\theta_s = \theta_m$ (rotation angle as mech one)
(NEW) (dq ref frame) → d align with PM NORTH

(dq) REF. FRAME (Dym. model)

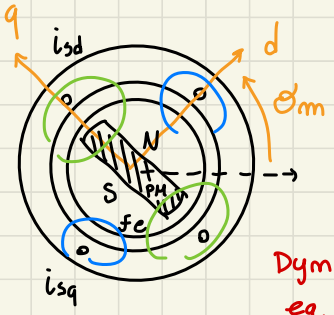
$$\begin{cases} \bar{V}_s = R_s \bar{I}_s + p \bar{\Psi}_s + J \dot{\theta}_m \bar{\Psi}_s \\ \Psi_{sd} = L_s i_{sd} + \Psi_{PM} \\ \Psi_{sq} = L_s i_{sq} \end{cases}$$

← motion term (new term due to movement)

only on d axis, taken as ROTOR reference

⇓ ④ Introduce anisotropy:

ANISOTROPY := ROTOR no more symm, property influenced by θ_m (ROTOR position)



In this machine design
go through
 $\begin{cases} i_{sd} \text{ lines} \rightarrow \text{larger air gap} \\ i_{sq} \text{ lines} \rightarrow \text{smaller air gap} \end{cases}$

$\Rightarrow L_q > L_d$
(with superposition principle, PM ~ AIR)

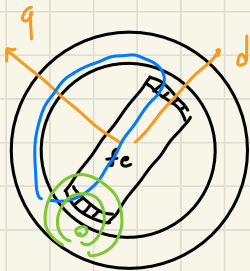
Dym eq.

$$\begin{cases} \bar{V}_s = R_s \bar{I}_s + p \bar{\Psi}_s + J \dot{\theta}_m \bar{\Psi}_s \\ \Psi_{sd} = L_d i_{sd} + \Psi_{PM} \\ \Psi_{sq} = L_q i_{sq} \end{cases}$$

$L_d \neq L_q$ IF ANISOTROPIC

Very similar Dym eq
(valid both for ISO/ANISOTROPIC)
($L_d = L_q = L_s$ ISOTROPIC)

With a different structure design of AC brushless machine:



$\begin{cases} L_d \text{ small air gap} \\ L_q \text{ large air gap} \end{cases} \rightarrow L_d > L_q \text{ on this design}$

↓
depending on the ROTOR design $L_d \geq L_q$
(comparison between dq inductance depends on ROTOR design)

⑤ then express TORQUE:

Through Energy Balance you can express torque

$$V_{sd} i_{sd} + V_{sq} i_{sq} = \text{Re} \{ \bar{V}_s \underline{i}_s \}$$

total power inside machine $\Rightarrow$

$$T_e = m_p \left[\underbrace{(L_d - L_q)}_{\substack{\text{1\% ISOTROPIC} \\ = 0}} i_{sd} i_{sq} + \psi_{PM} i_{sq} \right]$$

2 torque components
↓ due to ANISOTROPY and STATOR current

different approaches

• **IDEA 1** $\Leftarrow$ to solve machine (to reduce complexity)

↓
NEGLECT Reluctance torque (Anisotropy)

(rot of machine implemented this way)

$\rightarrow$ ISOTROPIC structure $\Rightarrow T_{\text{Reluctance}} = 0$

(or on IPM machine) $\rightarrow$ (you neglect OR NOT use $L_d - L_q$)

When we consider $L_d - L_q \sim 0$ side effect negligible...

to get ROBUST Rotor, PM inside! (side effect on mounting PM inside for a ROBUST machine $\rightarrow$ we neglect)

consider $i_{sd \text{ REF}} = 0$ (you JUST work on excitation torque by i_{sq} neglecting Reluctance torque)

While in INDUCTION machine you need i_{sd} $\rightarrow$ For $i_{sd} \rightarrow \psi_r$ to get a flux you need i_{sd}
 $\Rightarrow i_{sd} = \psi_r / M$ in steady state

While on AC Brushless ψ is given by (PM): no need current for flux! (don't need current to realize the flux, already realized by PM)

and you can consider

setting $i_{sd} = 0$ $\rightarrow$ you can use all

machine current is as $i_{sq} \rightarrow$ [torque current!]

use full current on q axis

$$i_{sq \text{ MAX}} = i_{s \text{ MAX}}$$

$i_{s \text{ MAX}}$
max current of machine, related to

- Joule losses
- Cooling system
- Service Duty limitations

(while in induction machine isq is not max!) : part of the is overall stator current should be used to control the flux ψ

↓
 $T_e = m_p \psi_{PM} i_{sq}$ AC Brushless (Imposing $i_{sd} = 0$)
 ↓ similar T_e expression

$[T_e = m_p \psi_r i_{sq}]$ Ind. machine FOC with 6 windings ← but IM very complex system, lots of leakage and effect

• I'm able to maintain this control strategy reference OR I have limitations?
 ↓ (maintain $i_{sd} = 0$ and control i_{sq})? to check if feasible, look @ STEADY STATE:

in steady state conditions (from dymeqim dq)

steady state with $i_{sd} = 0$

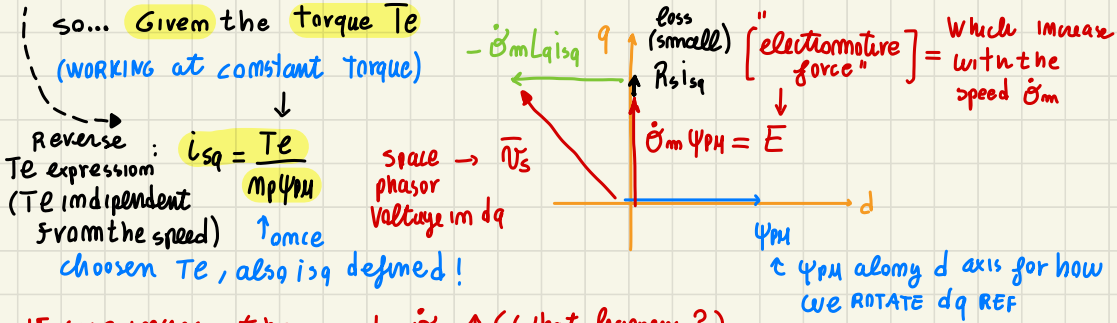
$$\begin{cases} V_{sd} = R_s i_{sd} + L_s \frac{di_{sd}}{dt} - \dot{\theta}_m L_q i_{sq} \\ V_{sq} = R_s i_{sq} + L_s \frac{di_{sq}}{dt} + \dot{\theta}_m L_d i_{sd} + \dot{\theta}_m \psi_{PM} \end{cases}$$

$$T_e = m_p \psi_{PM} i_{sq} \quad (\text{neglect Reluctance part})$$

(Im ROTOR P.O.V all quantities are constant!) (dq) REF. Looking from ROTOR, all constant
 $\dot{\theta}_m = \dot{\theta}_e = \omega_e$

$$\begin{cases} V_{sd} = -\dot{\theta}_m L_q i_{sq} \\ V_{sq} = R_s i_{sq} + \dot{\theta}_m \psi_{PM} \end{cases}$$

↓ With this REF. Frame (dq)



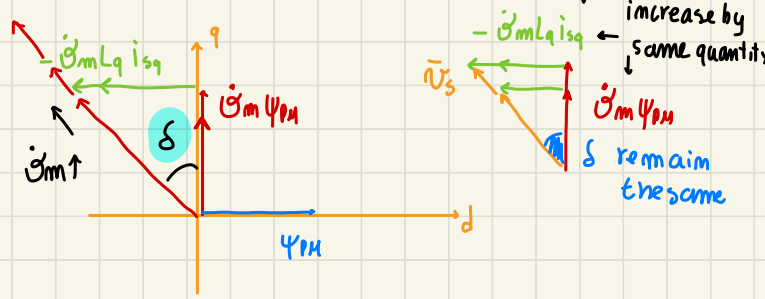
IF we increase the speed, $\dot{\theta}_m \uparrow$ (What happens?)

With medium- high speed: $R_s i_{sq} \ll \dot{\theta}_m \psi_{PM}$

constant
 $|-\dot{\theta}_m L_q i_{sq}| \uparrow$ increases!
 $|\dot{\theta}_m \psi_{PM}| \uparrow$

may be neglected (voltage drop on Resistance)

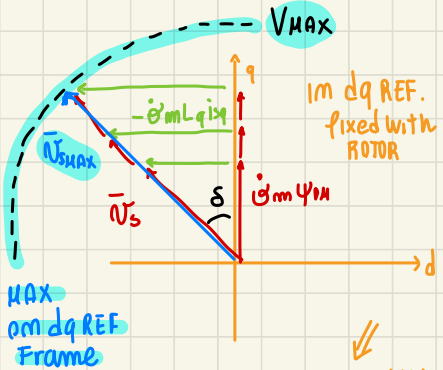
proportional to the speed $\dot{\theta}_m$! $\leftarrow \vec{V}_s$



IF $\dot{\theta}_m \uparrow$ the LOAD ANGLE δ does not change!

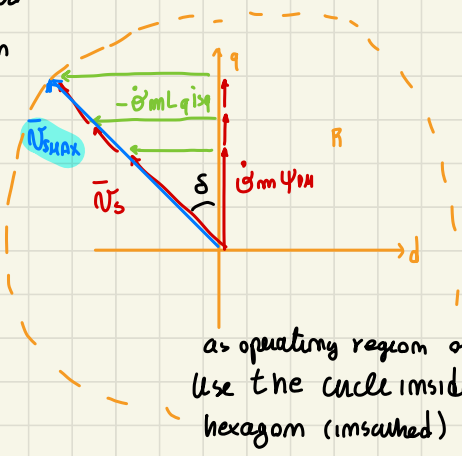
δ : LOAD ANGLE: it depends on T_e (isq) [once fixed T_e desired (so also isq) we have fixed δ]
 (increasing the speed ω_m , δ does not change!) (same triangular shape, given T_e)

$\Rightarrow$ at a certain point you reach $\bar{V}_{S_{MAX}}$ (you reach max feasible voltage)

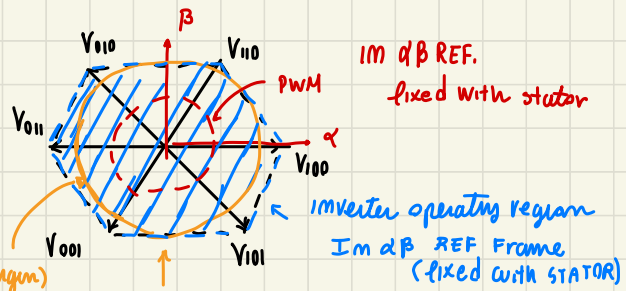


limit $\bar{V}_s$ using circle inscribed inside hexagon on (dq) axis

max VOLTAGE you can realize usually considering circle inscribed (with margin)



$\downarrow$ that limit express.
 VOLTAGE you can realize by means of space phasor by means of a suitable modulation technique:
 you can realize voltage inside inverter **operating region**



{ margin to check get the state estimation! }

When reaching $\bar{V}_{S_{MAX}}$ it corresponds to

$\Omega_m = \Omega_b$ BASE SPEED

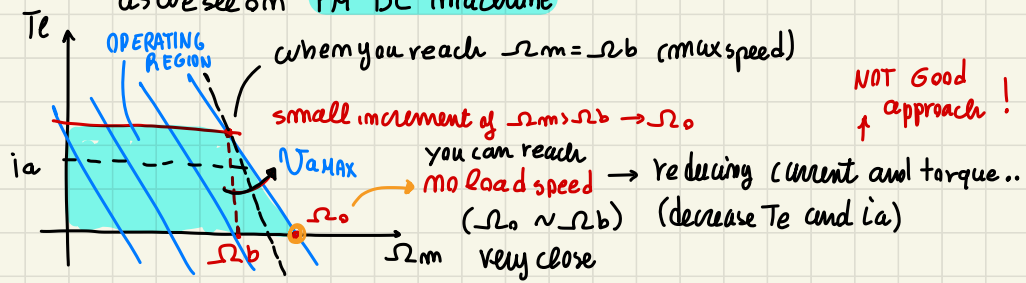
When you reach Max $\bar{V}_s$

$\downarrow$
 how to go above the Base speed?

$\Omega_m > \Omega_b \dots$

$\downarrow$ Go above Ω_b

as we see on **PM DC machine**

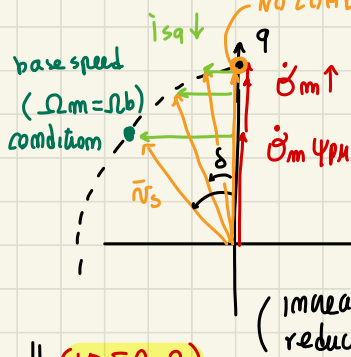


To solve this... in order to increase $\Omega_m > \Omega_b$, NEW IDEA!

om AC BRUSHLESS → (IDEA 1)

using a new strategy reducing isq

- NO LOAD condition for $i_{sq} = 0$ ($T_e = 0$)



isq ↓

(reduce current without maintaining same Torque $T \downarrow$)

I can increase $\Omega_m \uparrow$

reducing i_{sq} , so $\delta \downarrow$

- unter $i_{sq} = 0$, $\delta = 0 \Rightarrow T_e = 0 \rightarrow$

NO LOAD speed Ω_0

$$[\Omega_m \uparrow \text{ by } T_e \downarrow]$$

(similar to PM DC machine)

↓ (IDEA 2)

some years ago a company (siemens) propose a new solution to

obtain $\Omega_m > \Omega_b$

(gn above Ω_b in AC Brushless)

With a new RARE EARTH PM material • Ne Fe B_o

- Ne Fe B₀

- SaLo

for which the machine

can reach $\Omega_m > \Omega_b$

↓ using a proper control strategy

|| (ETR575) structure machine
to reach high Ω_m values

able to reach

$\Omega_m = 5 \Omega_b$! (a lot above Ω_b !)

↓ (working on flux weakening region)

If using $i_{sd} < 0$ (When reaching $\Omega_m > \Omega_b$)

for speed $> n_b$ introduce negative current:

$$[\dot{\theta}_m = m_p \Omega_m]$$

[isd $\neq 0$ model on (dq) axis]

As we can see on steady state dym eq.

for $i \leq d$ (neglecting $R_{i \leq d}$)

$$\int v_{sd} = - \dot{\theta}_m L_q i_{sq}$$

$$] \quad v_{sq} = +\dot{\theta} L_d i_{sd} + \dot{\theta} m \psi_{1H} \Rightarrow$$

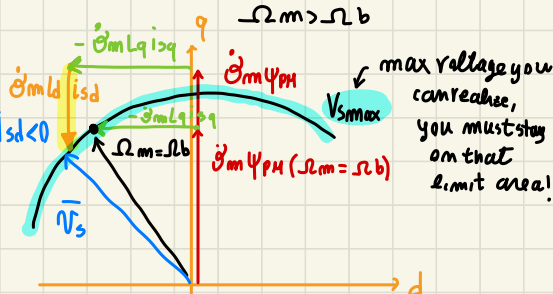
with $i_d < 0$

Diagram

changes

you can
go outside

the circle thanks to

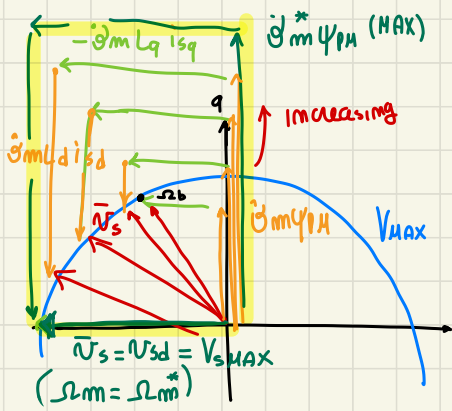
$$+ \dot{\theta}_m L_d \dot{i}_{sd}$$


you can go outside the circle by $isd < R$ term

ist ≤ 0 maintain $|\bar{U}_s| \leq V_{\max}$

Inside P.S OP. REGIONS (of inverter)

With a speed $\omega_m > \omega_b$ (reach higher speed than base speed!)



Realize working point for higher speed
(you are able to realize voltage reference $\vec{V}_s$ inside P.S operating regions also for $\Omega_m > \Omega_b$)

particular point:

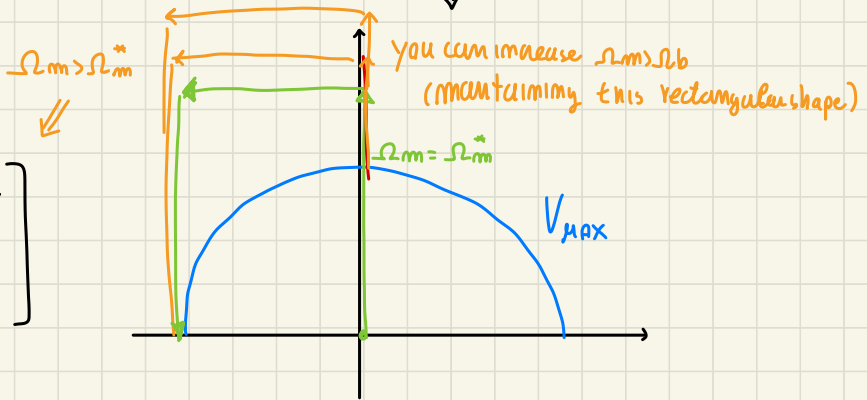
When you reach $V_{sd} = V_{MAX}$ (horizontal voltage, full $\alpha - d$ axis)
and $V_{sq} = 0$

point Ω_m^* correspond to that max speed Ω_m^*

relevant point of operating region!
(rectangular shape region)

(unique respect all other machine)

IF $\Omega_m > \Omega_m^*$

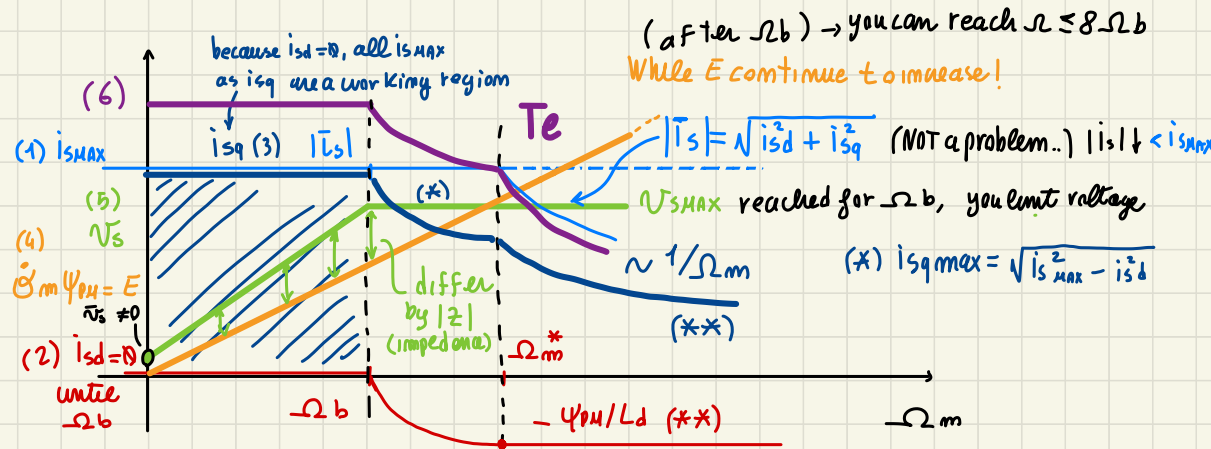


solutions when you need very high speed of the motor!

and you can reach $\Omega_m \approx 8 \Omega_b$
(low torque, high speed request)

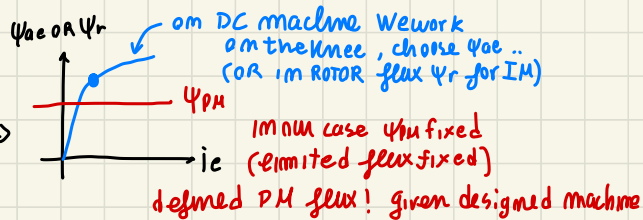
OPERATING REGIONS of AC BRUSHLESS

looking all quantities in steady state such that no damage on machine!



start from main limit:

- Limitations
- Toule losses (i_s)
 - max voltage due to Pow. supply ($\bar{V}_s$)
 - use properly Fe material (Ψ_{PM}) $\Rightarrow$

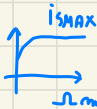


(1)

i_{sMAX} (Toule loss / source Duty / cooling syst)

→ more possible curves on datasheet depending on the machine design

→ (different fans, cooling ... $i_{sMAX}(\Omega_m)$ change according to machine)
(may be efficiency of cooling depends on Ω_m)



(2) We fix $i_{sd}=0$ until Ω_b then after you reach Base speed..

(3) $i_{sq} = i_{sMAX}$ until Ω_b , so V_{sm} such that $i_{sd}=0$ we can use full is limit on i_{sq} (i_{sq} AREA LIMIT! below i_{sq} you can stay)

(4) you have an emf component $E = \dot{\Psi}_{PM} \Psi_{PM}$ constant directly proportional to $\dot{\Psi}_{PM}$

(5)

Voltage $\bar{V}_s = R_s \bar{I}_s + j \dot{\Psi}_{PM} L_s \bar{I}_s + E \rightarrow \bar{V}_s > E$ by $(R_s \bar{I}_s + j \dot{\Psi}_{PM} L_s \bar{I}_s)$ $\bar{V}_s > E$

$E = j \dot{\Psi}_{PM} \Psi_{PM}$

With $\dot{\Psi}_{PM} \uparrow \bar{V}_s \gg E$

$|Z| \rightarrow$ difference between $\bar{V}_s, E$
impedance additional terms

STEADY STATE $\equiv (\dot{\Psi}_{PM} = \omega_m = \omega)$

condition

OPERATING Regions)

alternator
@ ω in steady-state

→ synchronous machine! In steady state speed mech = elect! (No slip!)

$V_s(\Omega_m = 0) \neq 0$ you have RSI's terms, voltage drop on resistance,
and then $\vec{V}_s \uparrow \uparrow$ with Ω_m

↳ you reach a LIMIT V_{sMAX} (due to P.S limitation)
LIMIT V_s

Instead even if V_{sMAX} limited $E = \dot{\Omega}_m \Psi_{PM} \uparrow \uparrow$ continue to increase
with Ω_m

↳ to CONTROL i_{sd} after Ω_b (siemens idea:)

for $\Omega > \Omega_b$, you move i_{sd} like a neg. curve

↳ introducing the effect discussed you use $i_{sd} < 0$
after Ω_b , until you reach Ω_m^* you decrease i_{sd} (neg.)

↳ (satisfy $|\vec{I}_s| = i_{sMAX}$ with $i_{sd}^2 + i_{sq}^2 = |\vec{I}_s|^2$)
because $i_{sd} \downarrow$

so also i_{sq} reduces with $i_{sd} \dots \rightarrow i_{sqMAX} = \sqrt{i_{sMAX}^2 - i_{sd}^2}$ maintain max modulus

Then:

for $\Omega_m > \Omega_m^*$: you should look at the limit vectors (Rectangular shape)

from this diagram... you need
to compare:

the two vertical and horizontal sides

↳ vertical sides

$$\dot{\Omega}_m L_d i_{sd} = -\dot{\Omega}_m \Psi_{PM}$$

↳ $i_{sd} = -\Psi_{PM}/L_d$ constant

i_{sd} for $\Omega_m > \Omega_m^*$

you no more decrease

negatively i_{sd} ! $i_{sd} = -\Psi_{PM}/L_d$ for $\Omega_m > \Omega_m^*$ (**)

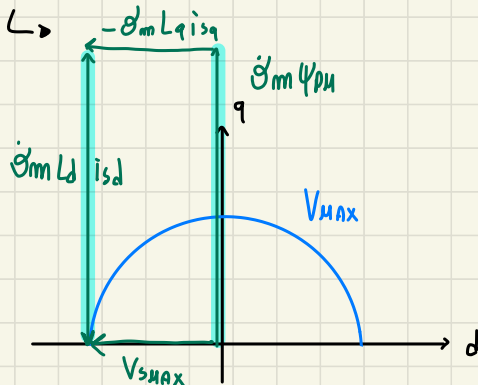
similarly, compare the horizontal side

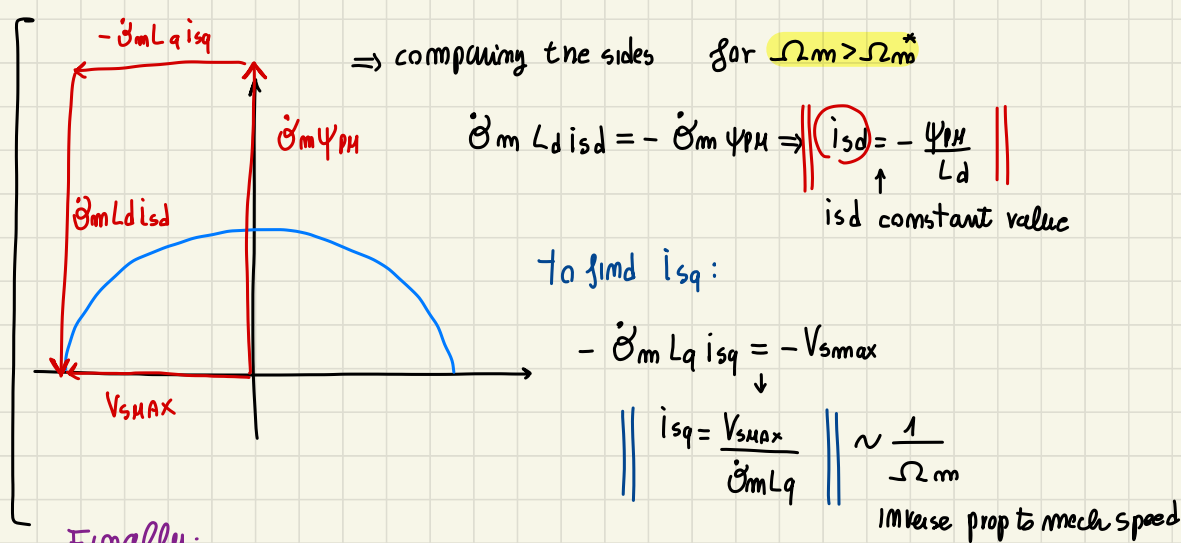
↳ $V_{sMAX} = +\dot{\Omega}_m L_q i_{sq}$ (same direction) $\rightarrow i_{sq} = \frac{V_{sMAX}}{\dot{\Omega}_m L_q} \sim \frac{1}{\Omega_m}$ for $\Omega_m > \Omega_m^*$ (**)

↳

so because $|\vec{I}_s| = \sqrt{i_{sd}^2 + i_{sq}^2}$ for $\Omega_m > \Omega_m^*$

because $i_{sd} \sim \frac{1}{\Omega_m}$ overall, also $|\vec{I}_s| \downarrow \sim \frac{1}{\Omega_m}$ (remain on $|\vec{I}_s| < i_{sMAX}$)
NO PROBLEMS!





Finally:

(6) $T_e \sim i_{sq}$ same shape of i_{sq} (proportional to i !)
 (last quantity to compute, after i_{sq})

↑
 we have deduced for **ISOTROPIC machine** [OPERATING REGION]

→ what if dealing with **ANISOTROPIC machine** (At the end we obtain $T_e \geq$ than $T_{e, ISOTROPIC}$ machine?)

on that case T_e , due to Reluctance torque $\begin{cases} i_{sd} < 0 \\ i_{sq} > 0 \end{cases}$

.. effect depends on

$L_d \geq L_q$ so

on ROTOR configuration

$$T_e = m_p \left[\underbrace{(L_d - L_q)}_{T_{REL}} i_{sd} i_{sq} + \psi_{PH} i_{sq} \right]$$

IF $L_d - L_q < 0 : T_{REL} < 0$ ($i_{sd} < 0$ for $\Omega_m > \Omega_m^*$) (Reluctance torque can be additive or subtractive!)

• on **IPH Robust rotor** if design to have $L_d < L_q$ you

can use **T_{REL} with $i_{sd} < 0 \rightarrow$ to get higher T_e on $\Omega_m > \Omega_m^*$** (or $\Omega_m > \Omega_b$ in general)

usage of **Reluctance torque** to gain T_e !

The **ANISOTROPY** has side effect, consider to control $i_{sd} = 0$, BUT when you reach Ω_b you use $i_{sd} < 0$:

Which guarantee a $T_e > 0$ is better to have **$L_d < L_q$**

(sometimes that **RELUCTANCE torque** $T_{REL} > 0$ can be used below Ω_b ...)

↓ NOT only on IPH:

(similar approach for **Synchronous Reluctance machine Assisted by PH**)
 (with $L_d < L_q$)

(as you can see $E \uparrow$ for $\Omega_m \uparrow$) $\downarrow$

after you reach $i_{sd} = -\psi_{PM}/L_d$, you cannot control more $i_{sd} \dots$

and when you reach $\Omega_m \simeq 8 \Omega_b$ (max limit)
on mechs speed



@ $8 \Omega_b :=$ When you switch off your control board, you cannot control more i_{sd} ($i_{sd} = 0$)



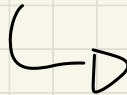
Immediately the voltage see the effect of $E = \dot{\Omega}_m \cdot \psi_{PM}$
Very high!

$\downarrow$ $E = 8 \Omega_b \psi_{PM}$ appears @ machine
Terminals! (no limit $\bar{V}_{smax}$)



All destroys $\rightarrow$ capacitor designed
around rated voltage: switches has a limit
 $\rightarrow$ breaking voltage on switches!

BREAK motor of the
control board switch off and
 $i_{sd} = 0 \rightarrow E$ applied on machine!
NO limit! Break all!



• IF $i_{sd} < 0$: $i_{sd} = -\psi_{PM}/L_d$ meg limit, you cannot control i_{sd} more.. and you reach $\Omega_m \approx 8 \Omega_b$ $\rightarrow$ when switch off @ $8 \Omega_b$, without control you have $i_{sd} = 0$

no way to reduce the voltage effect..

so E have had effect on capacitor! DESTROY MACHINE!

Attention to this Damage!

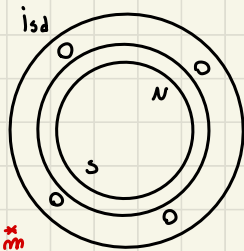
IF control board switch off with $\Omega_m \approx 8 \Omega_b$: all breaks!

Impact by means of i_{sd}

$$\begin{cases} \psi_{sd} = L_d i_{sd} + \psi_{PM} \\ \psi_{sq} = L_q i_{sq} \end{cases}$$

for $\Omega_m < \Omega_m^*$: $i_{sd} = 0$ $\psi_{sd} = \psi_{PM}$

than $i_{sd} < 0 \rightarrow \psi_{sd} = 0$ for $\Omega_m > \Omega_m^*$
 you are introducing on machine an i_{sd} current which works to reduce flux of permanent magnet



$$i_{sd} = -\psi_{PM}/L_d \quad (\Omega_m > \Omega_m^*)$$

means that

$$\psi_{sd} = L_d \left(-\frac{\psi_{PM}}{L_d} \right) + \psi_{PM} = 0$$

$\psi_{sd} = 0$ for $\Omega_m > \Omega_m^*$

thanks to this $i_{sd} < 0$

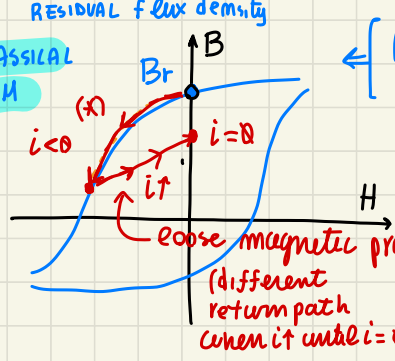
you get 0 flux emked! with sd coil

magnet and DE-MAGNETIZE the machine ROTOR

$\rightarrow$ introducing an electromotive force ($-\psi_{PM}$) opposite to the flux given by PM! to reduce emf!

SO: PROBLEM on Ω_m limit is related to the PM!

RESIDUAL flux density
 CLASSICAL PM



permanent magnet behaviour

(*) introducing negative current you move decreasing H than IF you again reduce $i=0$, $H \rightarrow 0$ you follow different path

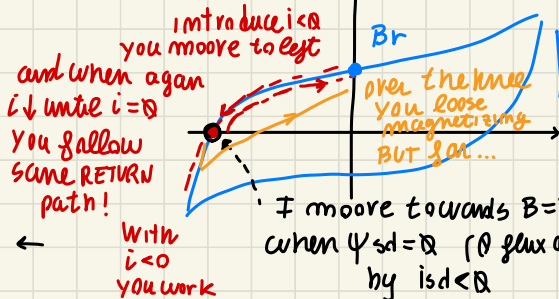
lose magnetic property! $\leftarrow$ you demagnetize your PM! definitely when you go after knee of $B(H)$

(IF you loose residual B_r you need to Remagnetize)

old material have problems of magnetization IF $\Omega > \Omega_b$!

The knee point is faster than the PM curve hysteresis!

with $i_{sd} \approx 0$ follow same hysteretic path.



while for different material

RARE EARTH PM

[has this hysteresis loop]

If move towards $B=0$ when $\psi_{sd}=0$ (0 flux density) by $i_{sd} < 0$

Good new material!

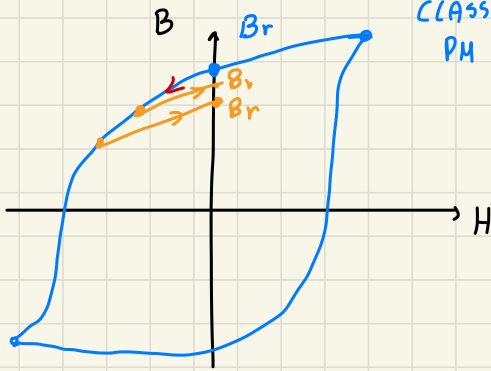
(carefully when move in short circuit, with high demagnetizing i)

and when $i_{sd} = 0$ you follow same RETURN path!

With $i < 0$ you work where flux density is 0

and when $i_{sd} = 0$ you have some B_r , don't DEMAGNETIZE!

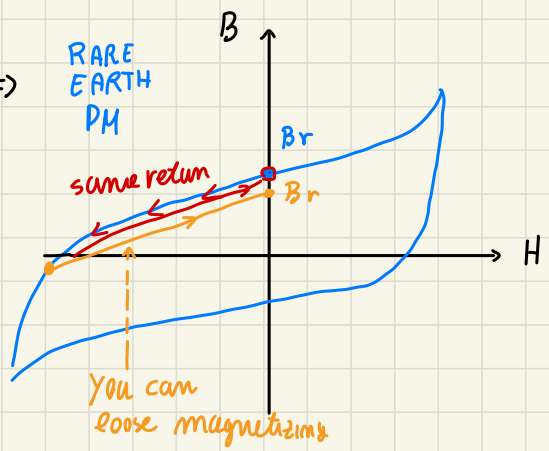
allow to go above Ω_b (decreasing i_{sd} a lot!)



CLASSICAL
PM

$\Rightarrow$

RARE
EARTH
PM



here $i_s d < 0$ very limited,
you loose quickly
residual flux!

(you cannot
go above Ω_b)!
cannot use $i_s d < 0$
strategy

BUT allow you to go a lot
above $\Omega_m > \Omega_b$! apply $i_s d < 0$
and use previously described
strategy $\Omega_m > \Omega_m^* \dots i_s d < 0$

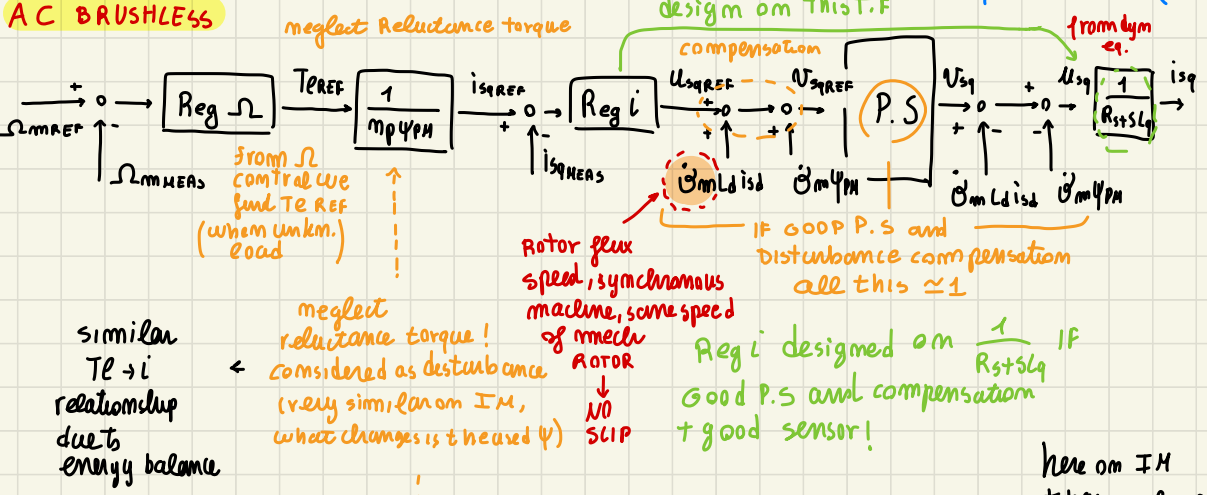
AC BRUSHLESS CONTROL SCHEME

(from dym eq.)
MODEL

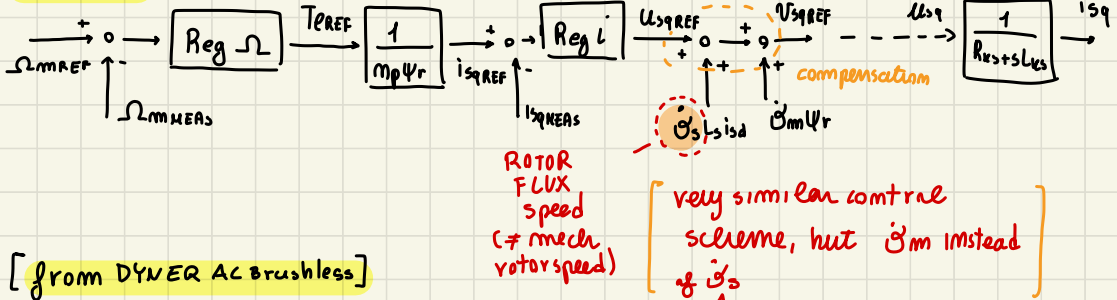
↓
Respect the previously studied control.. [IM (induction machine)]

SPEED CONTROL

AC BRUSHLESS



IM RFOC



[from DYMER AC Brushless]

$$\begin{cases} V_{sd} = R_s i_{sd} + L_d \frac{di_{sd}}{dt} - \Omega_m L_q i_{sq} \\ V_{sq} = R_s i_{sq} + L_q \frac{di_{sq}}{dt} + \Omega_m L_d i_{sd} + \Omega_m \psi_m \end{cases}$$

$\underbrace{\hspace{10em}}_{u_{sq}} \quad \underbrace{\hspace{10em}}_{\text{compensation}}$

↑
this is the speed control...

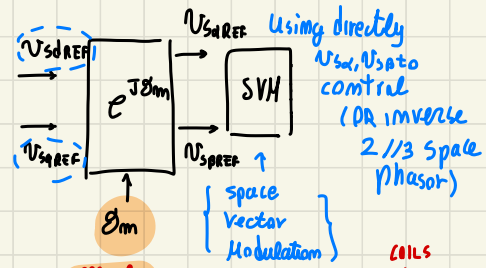
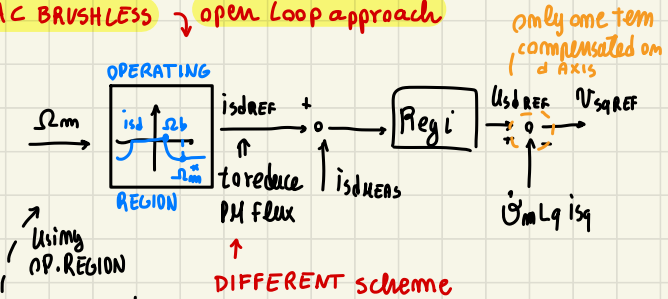
We can do the same for the FLUX control $\Rightarrow$



FLUX CONTROL (Great difference between the two..)

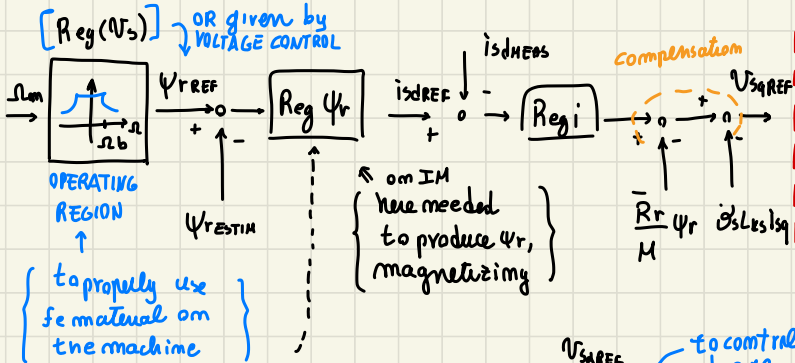
to move to real quantity

AC BRUSHLESS → open Loop approach

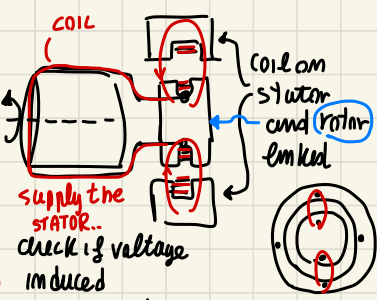


Both need to control i_d , BUT on IM you need to use magnetizing current to produce ψ_r ! while on AC Brush you reduce i_d when you need to reduce PM flux (needed for LOW Ω_m)

IM RFOC = open loop cheaper / flux controller



made out of 2 machines transformer and coils..



connected to a Rotating coil linkage by $\downarrow$

$\sin(\theta_m), \cos(\theta_m)$

can be measured by RESOLVER

here on induction machine, you need to know

$\theta_s, \dot{\theta}_s, \psi_r$ by observer / estimator,

LOTS calculation needed

Algorithm to estimate rotor flux modulus, position, speed ...

↓ similar CONTROL SCHEME

but on IM you need estimator

algorithm i_s, V_i , simpler on AC Brushless...

While on IM, lots more info needed

$e^{j\theta_m} = \cos(\theta_m) + j\sin(\theta_m)$

Resolver = 2 machine in series able to give the $\sin, \cos$ of θ_m (or other time using an ENCODER)

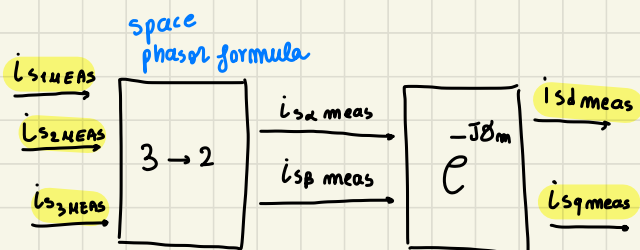
↓

info about $\theta_m, \dot{\theta}_m$ by sensor on motor shaft in AC Brushless (simple to compute!)

also IF you need... **another PROBLEM Related to:**

IF you have $i_{s1,2,3}$ measured to go on dq REF.?

δ_m necessary also for this



$\uparrow \delta_m$ ← angle δ_m used also for this transformation

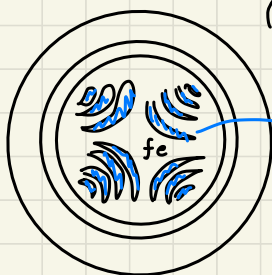
(same on induction machine BUT with θ_s instead of δ_m): $e^{-j\theta_s}$ on IM

AC BRUSHLESS Without permanent magnet

SYNCHRONOUS RELUCTANCE MACHINE ("SYNCHREL")

(same approach of classical AC Brushless) → START with same reasoning of AC brushless

EXAMPLE
(4 poles) machine with Barrier inside
(sheet of Fe material with AIR)



on that machine we start with the same approach of AC Brushless, considering

- the **SMPM** (machine with (PM))
(ideally START from PM ISOTROPIC!)

from SMPM dyn. equation:

goto REF

($\alpha \beta$)

- move to new REF

(dq) and **ANISOTROPY** included

- $$\begin{aligned} \bar{V}_{s1} &= R_s \bar{i}_{s1} + p \bar{\psi}_{s1} \\ &\vdots \\ \bar{\psi}_{s1} &= \dots \end{aligned}$$

- $$\bar{V}_{s\alpha\beta} \Rightarrow \begin{cases} \bar{V}_s = R_s \bar{i}_s + p \bar{\psi}_s + j \dot{\delta}_m \bar{\psi}_s \\ \bar{\psi}_{sd} = \bar{L}_d \bar{i}_{sd} + \bar{\psi}_{PM} \\ \bar{\psi}_{sq} = \bar{L}_q \bar{i}_{sq} \end{cases}$$

- and torque expression

$$T_e = m_p [(L_d - L_q) i_{sd} i_{sq} + \psi_{PM} i_{sq}]$$

$\uparrow L_d \neq L_q$ anisotropy

from that... we introduce **SynchRel simplification...**



SynchRel simplification

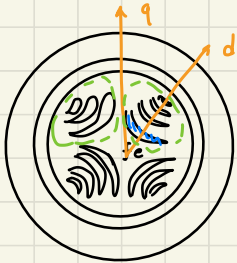
$\psi_{PM} = 0$ there is NO permanent magnet!

$$\begin{cases} \bar{V}_s = R_s \bar{I}_s + p \bar{\psi}_s + j \omega_m \bar{\psi}_s \\ \psi_{sd} = L_d i_{sd} + \cancel{\psi_{PM}} \\ \psi_{sq} = L_q i_{sq} \end{cases}$$

only Reluctance torque

$$T_e = m_p \left[(L_d - L_q) i_{sd} i_{sq} \right] \quad (\text{no excitation torque!})$$

this machine design try to obtain great $L_d - L_q$ difference! → due to barrier design of AIR inside... properly designed ROTOR air hole..



$L_d \ll L_q$ with this design,

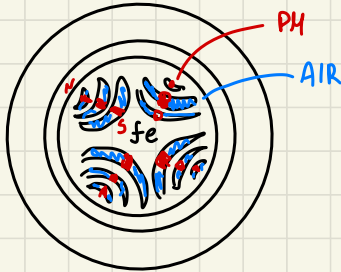
because d axis lines go through lot of air! while q through lot of σ_c , small air gap so bigger L_q

STRATEGY! ⇒ good Rotor shape to increase L_d, L_q difference, like $\left\| \frac{L_q}{L_d} \approx 9 \right\|$

With SYNCHREL assisted by PM

used on ELECT CAR, WITH BOTH RELUCTANCE + EXC. TORQUE

it has some PM inside which helps to increase the efficiency of the machine..



Model is equal to IPM OR ANISOTROPIC Machine

(same approach as before!) (to find final model)

idea to realize great difference between L_d and L_q

↑ higher cost! → allow to use both torque components

↓

PROBLEM of this machine

nobody want to produce! → opening the machine, it is easy to reproduce, the idea of design cannot be protected by copyright (internal design easy to reproduce if opened..)

CONTROL SCHEME NOT an unique solution!

$$(\tau = L/R \text{ with same } R...)$$

some IDEAS

- for example, looking @ time constant, being $L_d \ll L_q \Rightarrow \tau_d \ll \tau_q$

→ to properly use material fe

so is better to use i_{sq} constant (slower)

better response time of system torque

and control torque T_e by i_{sd} (which is faster)

@ $\omega_m = \omega_b$

When reaching max speed...

$$\begin{cases} V_{sd} = R_s i_{sd} + L_d p i_{sd} - \dot{\theta}_m L_q i_{sq} \\ V_{sq} = R_s i_{sq} + L_q p i_{sq} + \dot{\theta}_m L_d i_{sd} \end{cases}$$

increase with $\dot{\theta}_m$

with i_{sq} constant
this increase with $\dot{\theta}_m$,
at one point reached $\dot{\theta}_{mmax}$,
you reach V_{sdmax} !

@ this point you need

to reduce $i_{sq} \Rightarrow$ Demagnetizing FLUX WEAKENING REGION

in some research

we try to maximize:

$$\frac{T_e}{|i_s|}$$

approach:

Maximum Torque per Ampere [MTPA]

↓

{ I try to find the operation point: values i_{sd}, i_{sq}
to maximize T_e ! or given T_e such to minimize $|i_s|$ }
so to have minimum values of Joule losses
→ by some TABLE of optimization

more possible theory to control the SYNCHREL machine!

→ look the efficiency or time constant τ to properly design control for SynchRel ...

Summary

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6. AC brushless

6.1 Introduction

The expression AC brushless drives is used to indicate those drives that use a synchronous machine with permanent magnets: the excitation magnetic field, in such machines, is provided by permanent magnets.

The scope of these drives is mainly limited to low power (typically less than 50 kW) for the limitations imposed by the magnetic materials currently available to the construction of machines: the size and cost of the magnets are very high. However, this is an important field of application and in significant growth, including numerically controlled machine tools, industrial automation, robotics, light traction, heavy traction, wind generation. Furthermore, due to the virtual absence of rotor losses these machines do not require forced ventilation and are therefore suitable for applications like aerospace or contaminated environments.

The basic scheme of a servo drive (see Figure 6-1) consists of: a permanent magnet synchronous motor, a static converter (in this case a two-stage converter consists of a diode (or a IGBT/MOSFET) bridge rectifier and an inverter), a position sensor, some current and voltage sensors and a control device that operates based on information provided by the sensors.

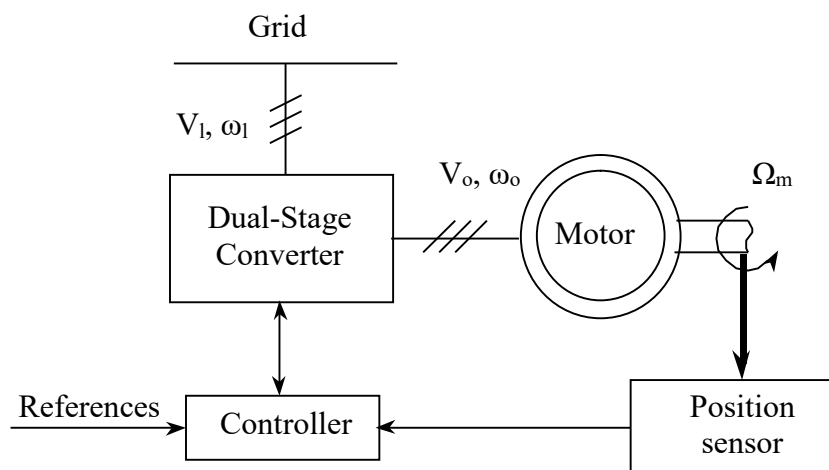


Figure 6-1: AC brushless drive scheme

The following considerations apply, mainly, to drives for axes of machine tools of small and medium power (2-10kW), but are also valid for high power drives.

6.2 General characteristics of the drive

In the field of drives for machine tools, a distinction is often introduced between the axes and spindles, where the first are intended solely for the motion control, while the second ones are used for rotation of the piece.

The next step is to analyze briefly the requirements of an axis drive: its task is essentially to move a tool or a piece, accelerating and decelerating from zero to a given speed, by means of a suitable and controlled torque: this task must be carried out in accordance with important requirements:

- Total bidirectionality drive with practically no dead zone around zero speed, in both static and dynamic conditions;
- The ratio between minimum and maximum speeds should be greater than 10 (at rated torque) and, passing from zero to rated load, the speed must not drop more than 1/10 of the maximum speed.

These characteristics can be summarized by saying that the drive must have a high static stiffness, meaning the static stiffness equal to the ratio between the applied external torque and the produced speed variation.

Another key element of technical opinion is the bandwidth of the speed control loop: a high performance axis drive requires a bandwidth regarding the phase not less than 40 Hz (ie the frequency at which the phase delay exceeds 45° above 40 Hz) while regarding the amplitude (frequency at which there is an attenuation of 3 dB) is not less than 70 Hz. These characteristics are valid in operation around zero speed, but they must stay within a margin of 20% also in the case of an "inertial load" (defined conventionally as a load equal to half the inertia of the machine) or of a "rated torque load" (conventionally defined as a torque equal to one half of the rated one). The magnitude of the bandwidth of the speed control loop can therefore be taken as an index of dynamic stiffness of the drive.

Finally, the acceleration provided by the machine in the presence of an inertial load or torque as defined above must be elevated, approximately, to 100 rad/s^2 ; however the maximum speeds required are not generally very high, about 200 rad/s.

The purpose of these requirements, however, closely related to each other, is that you can get, with the help of rigid kinematic chains, a very accurate position control in order to achieve a high degree of finish of the workpiece.

6.3 Synchronous machine model

In the following, the mathematical model of the synchronous machine is briefly introduced. This model has implied some simplifications regarding the constructive nature of the machine, but it appears to be sufficiently adequate to its control.

Consider the machine shown schematically in Figure 6-2

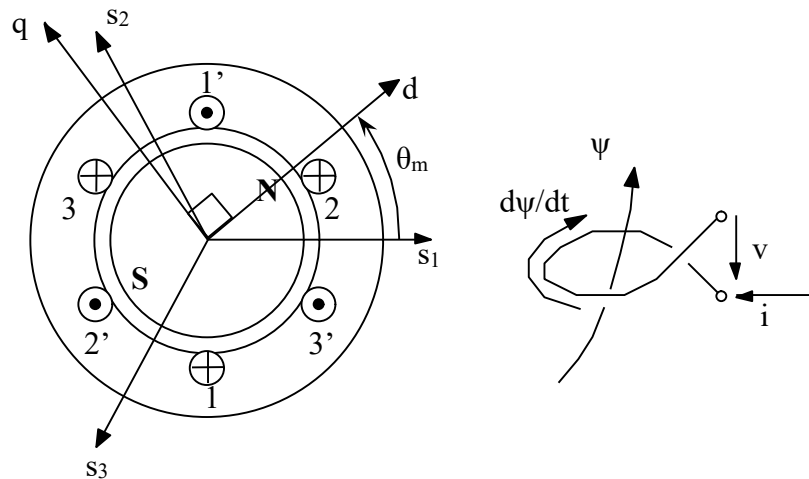


Figure 6-2: Schematic diagram of the brushless motor

It consists of a cylindrical stator with a symmetrical three-phase winding so as to generate a distribution of sinusoidal magnetomotive force in the air-gap, and a rotor flux created by permanent magnets (for the windings apply the conventions of Figure 6-2 and without damper cages. We also admits that the stator and the rotor are made of laminated material with infinite permeability. Saturation, hysteresis of iron and anisotropy of the machine due to the slots are neglected.

There are two main solutions for the rotor: with permanent magnets glued to the rotor surface (SMPM Surface Mounted Permanent Magnet, Figure 6-3) and interior magnets (IPM Interior Permanent Magnet, Figure 6-4).

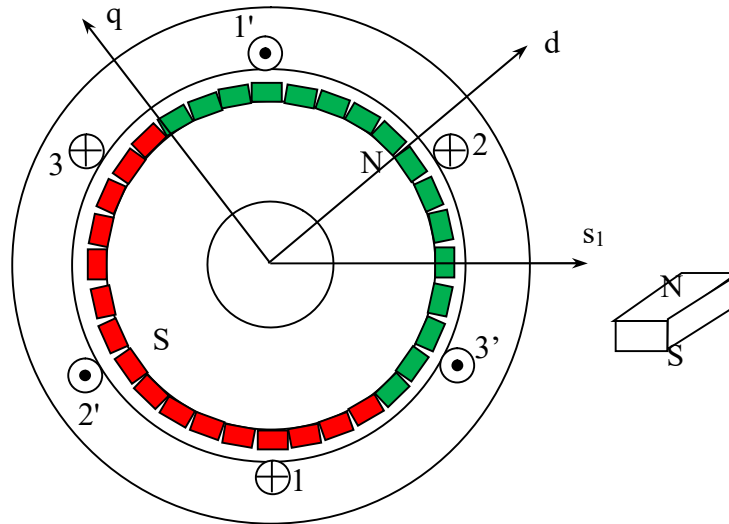


Figure 6-3: Example of a SMPM synchronous machine, 2 poles (green magnets have North outward while the South towards the rotor; red ones vice versa)

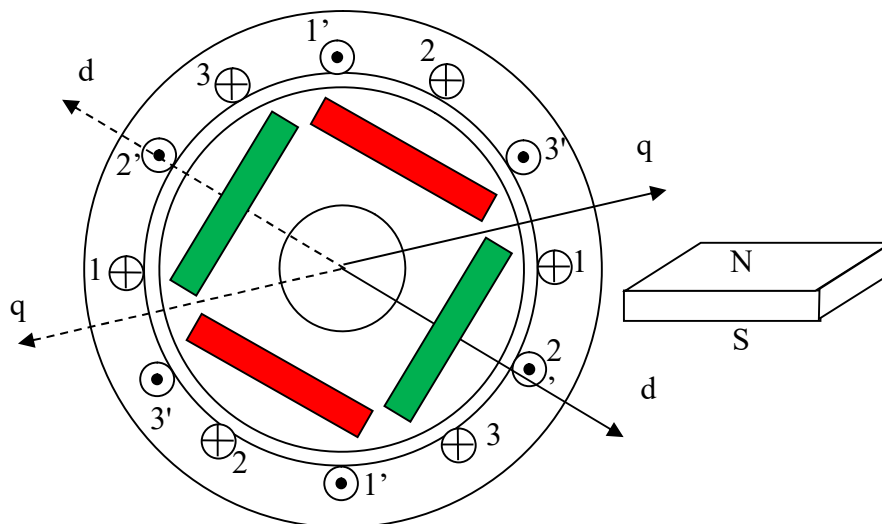


Figure 6-4: Example of IPM synchronous machine, 4 poles (green magnets have North outward while the South towards the rotor; red ones vice versa)

In the first case the machine can be considered isotropic (the self and mutual inductances do not depend on the angular position); in the second case the machine is anisotropic.

Starting from the stator equations and the current-flux relationship (for which, given the above assumptions, the superposition principle is valid)

$$v_{s1} = R_s i_{s1} + p \psi_{s1}$$

$$v_{s2} = R_s i_{s2} + p \psi_{s2}$$

$$v_{s3} = R_s i_{s3} + p \psi_{s3}$$

$$\psi_{s1} = L_{ss}(\theta_m) i_{s1} + M_{ss}(\theta_m) i_{s2} + M_{ss}(\theta_m - \frac{2}{3}\pi) i_{s3} + \psi_{pm}(\theta_m)$$

$$\psi_{s2} = L_{ss}(\theta_m - \frac{2}{3}\pi) i_{s2} + M_{ss}(\theta_m) i_{s1} + M_{ss}(\theta_m + \frac{2}{3}\pi) i_{s3} + \psi_{pm}(\theta_m - \frac{2}{3}\pi)$$

$$\psi_{s3} = L_{ss}(\theta_m - \frac{4}{3}\pi) i_{s3} + M_{ss}(\theta_m - \frac{2}{3}\pi) i_{s1} + M_{ss}(\theta_m + \frac{2}{3}\pi) i_{s2} + \psi_{pm}(\theta_m - \frac{4}{3}\pi)$$

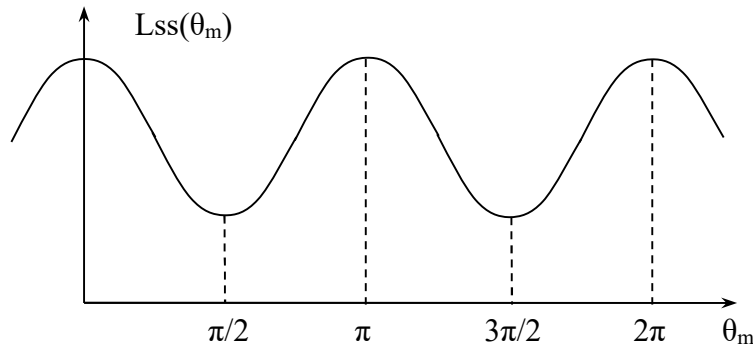


Figure 6-5: Self inductance behaviour of the first winding

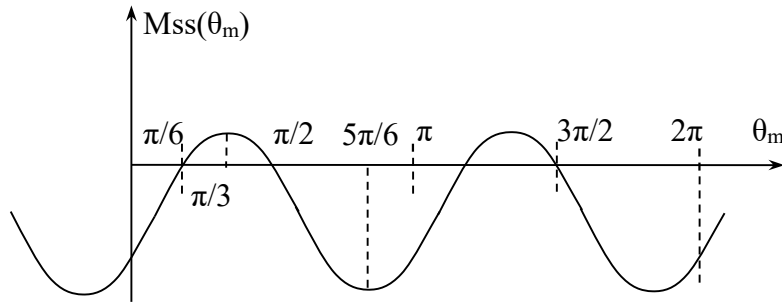


Figure 6-6: Behaviour of the mutual inductance between the first and the second winding

Consider now, for the sake of simplicity, an isotropic machine (the case of anisotropic machine will be taken up after). Recalling that the sum of the three phase currents is zero due to the connection (in isolated star or delta connection), we have:

$$\psi_{s1} = L_s i_{s1} + \psi_{pm}(\theta_m)$$

$$\psi_{s2} = L_s i_{s2} + \psi_{pm}(\theta_m - \frac{2}{3}\pi)$$

$$\psi_{s3} = L_s i_{s3} + \psi_{pm}(\theta_m - \frac{4}{3}\pi)$$

where $L_s = L_{ss} - M_{ss}$ (synchronous inductance).

The synchronous PM machine (AC brushless) is characterized by a flux linked with the stator windings, due to the permanent magnets with a sinusoidal waveform (due to a suitable distribution of the windings in the stator slots and/or a shaping of the magnets).

$$\psi_{pm}(\theta_m) = \hat{\psi}_{pm} \cos(\theta_m)$$

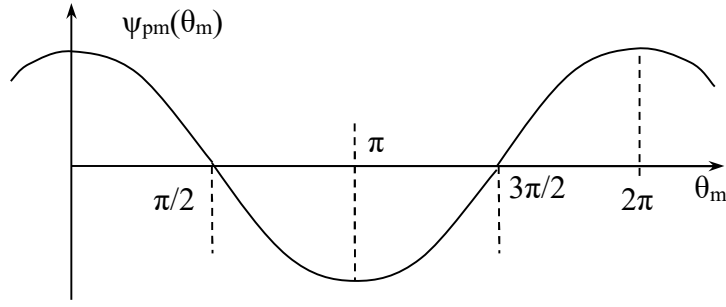


Figure 6-7: Waveform of the flux linked with the stator windings due to the PM

Applying now the space phasor formula in a stationary reference frame (α axis has the same direction of the magnetic axis of the first winding), we obtain the following relationships:

$$\begin{aligned}\bar{v}_s^{\alpha\beta} &= R_s \bar{i}_s^{\alpha\beta} + p \bar{\psi}_s^{\alpha\beta} \\ \bar{\psi}_s^{\alpha\beta} &= L_s \bar{i}_s^{\alpha\beta} + \sqrt{\frac{3}{2}} \hat{\psi}_{pm} e^{j\theta_m}\end{aligned}$$

This procedure has transformed the three windings machine into a two-phase machine, equipped with two windings, fixed with the stator and orthogonal each other.

Defined ψ_{pm} as

$$\psi_{pm} = \sqrt{\frac{3}{2}} \hat{\psi}_{pm}$$

and passing to a reference frame fixed with the PM flux (axis " d " fixed with the North of the permanent magnets), we obtain the following relations ($\bar{F}^{\alpha\beta} = \bar{F}^{dq} e^{j\theta_m}$)

$$\begin{aligned}\bar{v}_s &= R_s \bar{i}_s + p \bar{\psi}_s + j \dot{\theta}_m \bar{\psi}_s \\ \psi_{sd} &= L_s i_{sd} + \psi_{pm} \\ \psi_{sq} &= L_s i_{sq}\end{aligned}$$

where $\bar{v}_s = v_d + jv_q$ and v_d, v_q are the projections of the voltage space phasor on the axes d and q (fixed with rotor).

This procedure has transformed the two-phase machine, equipped with two windings, fixed with the stator and orthogonal each other, into a two-phase machine, with two windings equally distributed, whose magnetic axes (orthogonal each other) are moving fixed with the d and q axes.

If the machine is anisotropic, the winding sd , with magnetic axis in the direction d , will be crossed by the current i_{sd} and will support the flux ψ_{sd} through a self inductance L_d that will be, now, constant (no more function of the mechanical angle) and different from the self inductance L_q of the winding sq .

$$\begin{aligned}\bar{v}_s &= R_s \bar{i}_s + p \bar{\psi}_s + j \dot{\theta}_m \bar{\psi}_s \\ \psi_{sd} &= L_d i_{sd} + \psi_{pm} \\ \psi_{sq} &= L_q i_{sq}\end{aligned}$$

In the case of Figure 6-4, since the permanent magnet, by applying the principle of superposition and considering only the effect of the currents, has a behaviour similar to air, the flux along the direct axis ψ_{sd} must pass through more air than the quadrature axis: in this case is $L_d < L_q$.

The energy balance says that:

$$\text{Re}(\bar{v}_s \underline{i}_s) = R_s \underline{i}_s^2 + \text{Re}(\underline{i}_s p \bar{\psi}_s) + \text{Re}(j \dot{\theta}_m \bar{\psi}_s \underline{i}_s)$$

The term at the left is the total power entering the machine, the first term on the right is the power losses (due to Joule effect), the second term is the variation of the internal energy (magnetic energy stored in the inductances L_d e L_q : $i_{sd} p \psi_{sd}$ e $i_{sq} p \psi_{sq}$), while the third is the mechanical power.

$$\begin{aligned} P_m &= \text{Re}(j \dot{\theta}_m \bar{\psi}_s \underline{i}_s) = -\dot{\theta}_m \text{Im}(\bar{\psi}_s \underline{i}_s) = -\dot{\theta}_m \text{Im}[(L_d i_{sd} + \psi_{pm} + j L_q i_{sq})(i_{sd} - j i_{sq})] = \\ &= \dot{\theta}_m [(L_d - L_q) i_{sd} i_{sq} + \psi_{pm} i_{sq}] \end{aligned}$$

The expression of the torque is:

$$T_e = \frac{P_m}{\Omega_m} = \frac{P_m}{\Omega_m / n_p} = n_p [(L_d - L_q) i_{sd} i_{sq} + \psi_{pm} i_{sq}]$$

The first term is called "reluctance torque" (null if the machine is isotropic) and the second one "excitation torque".

Finally, there is the equation of the mechanical energy balance:

$$p \dot{\theta}_m = \frac{n_p}{J} (T_e - T_r)$$

6.4 Vector control

The vector control of synchronous machine is based on a suitable choice of reference axes d and q , used by the controller of the power converter so that a component of the space phasor of the stator current acts only on the flux while the other one on the electromagnetic torque (in the air-gap). In this way, the synchronous motor is controlled like a DC machine where the regulator acts separately on the excitation current (flux) and the armature current (torque).

To illustrate how the vector control of synchronous machine, recall the above mentioned model of the machine:

$$\bar{v}_s = R_s \bar{i}_s + p \bar{\psi}_s + j \dot{\theta}_m \bar{\psi}_s$$

$$\psi_{sd} = L_d i_{sd} + \psi_{pm}$$

$$\psi_{sq} = L_q i_{sq}$$

$$p \dot{\theta}_m = \frac{n_p}{J} (T_e - T_r)$$

$$T_e = n_p \text{Im}(\bar{i}_s \underline{\psi}_s) = n_p ((L_d - L_q) i_{sd} i_{sq} + \psi_{pm} i_{sq})$$

Substituting the stator flux components with their dependence on the stator currents and the permanent magnets flux (with the assumption that the machine parameters are time invariant) we have:

$$\begin{aligned}v_{sd} &= R_s i_{sd} + L_d p i_{sd} - \dot{\theta}_m L_q i_{sq} \\v_{sq} &= R_s i_{sq} + L_q p i_{sq} + \dot{\theta}_m \psi_{pm} + \dot{\theta}_m L_d i_{sd}\end{aligned}$$

and finally:

$$T_e = n_p [(L_d - L_q) i_{sd} i_{sq} + \psi_{pm} i_{sq}]$$

6.5 Operating regions

By analyzing the expression of the torque you can see that the anisotropy of the rotor, represented by the difference $(L_d - L_q)$, produces a torque of anisotropy (or "reluctance torque"); in the following, unless otherwise specified, we will consider only the isotropic case of synchronous machine, so it results $L_d = L_q = L_s$.

In this case we have:

$$T_e = n_p \psi_{pm} i_{sq}$$

Looking at the last expression, it is evident therefore that, for the torque birth, only the quadrature component of stator current is effective, while direct component has no effect on the torque. So in order to minimize the magnitude of the current space phasor (and consequently the losses), vector control should operate on the power converter in such a way that is, at any time:

$$i_{sd} = 0$$

From the point of view of the torque control, therefore, the behaviour of a synchronous motor with vector control is similar to that of a DC machine: in this analogy, the quadrature component of stator current assumes the role of the armature current. Unlike a DC machine, however, with an i_{sd} equal to zero, it is not possible to act on the excitation (weakening) because the excitation field is provided by permanent magnets.

Of course, in this case, i_{sd} and i_{sq} are currents flowing in fictitious windings (not real windings), but they are the components, on a reference frame fixed with rotor, of a current space phasor created by a three-phase system of currents: the transition from one system to the other is obtained by the space phasor formulas.

It is clear now the important function performed by the position sensor that appears in Figure 6-1, as it is the device that provides, to the control system, the instant value of the angular position of the rotor (and therefore the position of the permanent magnets flux ψ_{pm}), needed to realize the above mentioned transformation.

In steady state operation and with the machine powered by a three-phase symmetrical direct voltage (with a pulsating ω frequency), the mechanical speed $\dot{\theta}_m$ (also indicated by ω_m) coincides with the frequency ω . In the reference frame fixed with the rotor, therefore, the voltage phasor is stationary and constant: accordingly, all the electric quantities are constant and it results (i_{sd} is set to 0 and the derivatives are null):

$$\begin{aligned}\dot{\theta}_m &= \omega_m = \omega \\v_{sq} &= R_s i_{sq} + \omega_m \psi_{pm} = R_s i_{sq} + E \\v_{sd} &= -\omega_m L_s i_{sq}\end{aligned}$$

One can thus draw the following vector diagram:

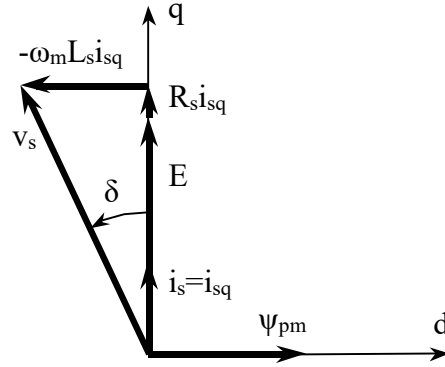


Figure 6-8: steady state vector diagram of an AC brushless

In the case in which the voltage drop on the resistance R_s is negligible with respect to E , it is noted that, during a the variation of the mechanical speed, the angle δ remains constant (both catheti are proportional to the mechanical speed) and that the amplitude of the voltage vector increases linearly with the same speed. The speed to which the vector voltage reaches the maximum value given by the power supply (compatibly with the degree of insulation of the stator windings) is called base speed ω_{base} . This limit, in the case of three-phase inverter supply, is represented by a circle whose radius is slightly smaller (necessary to maintain a voltage margin for the dynamic control of the current) than the radius of the circle inscribed in the hexagon of the field of operation of the inverter.

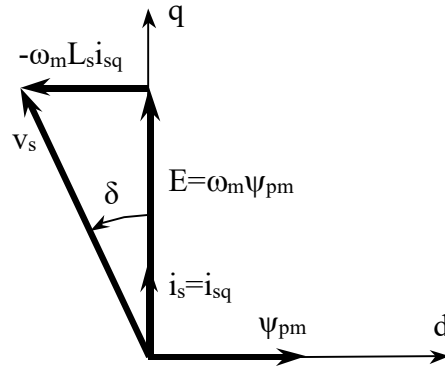


Figure 6-9: steady state vector diagram of an AC brushless (neglecting the resistive drop)

Suppose now that the machine is fed so as to produce a negative direct axis component of the current; in this case $i_{sd} < 0$ (no more $i_{sd} = 0$); then, at steady state, the following relations are valid:

$$v_{sd} = R_s i_{sd} - \omega_m L_s i_{sq}$$

$$v_{sq} = R_s i_{sq} + \omega_m L_s i_{sd} + E$$

which implies the following diagram (neglecting the resistive drop):

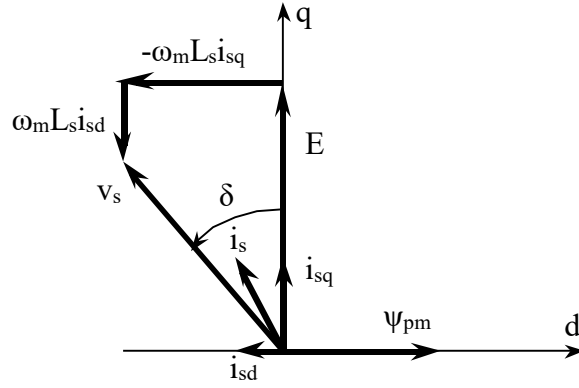


Figure 6-10: steady state vector diagram of an AC brushless when $i_{sd} < 0$

From the diagram, it is evident that, for the same value of i_{sq} (and then the torque), the addition of a negative component i_{sd} led to a reduction of the stator voltage and an increase of the stator current. Having to keep the stator current within a predetermined maximum value imposed by the characteristics of the components of the power supply and by thermal effects, thus leads to the need to reduce the i_{sq} (ie the torque) when i_{sd} increases in its absolute value.

These observations suggest the following procedure:

- for speeds below the base one (ω_{base}), it is suitable to put $i_{sd} = 0$ and the component i_{sq} equal to the maximum allowed by the thermal constraints (i_{max}) so to operate in a region with constant torque, equal to the maximum possible (Figure 6-11 (a)); increasing supply frequency ω , the emf E and the stator voltage increase and therefore the power converter has to supply this voltage. Once at the base speed, the voltage is equal to the maximum voltage supplied from power converter except for a certain margin necessary to control the current (Figure 6-11 (b));
- for speeds higher than the base one, it is not possible to maintain $i_{sq} = i_{max}$ because the further increase of E would require a stator voltage higher than the maximum available by the power converter: in this region, therefore, i_{sq} has to be reduced to leave place to a negative value of i_{sd} in order to validate the following inequality:

$$i_s = \sqrt{i_{sd}^2 + i_{sq}^2} \leq i_{max}$$

Doing so it is verified that $v_s = v_{max}$ e $i_s \leq i_{max}$, but with an i_{sq} (and then the torque) less than in cases where it is $i_{sd} = 0$.

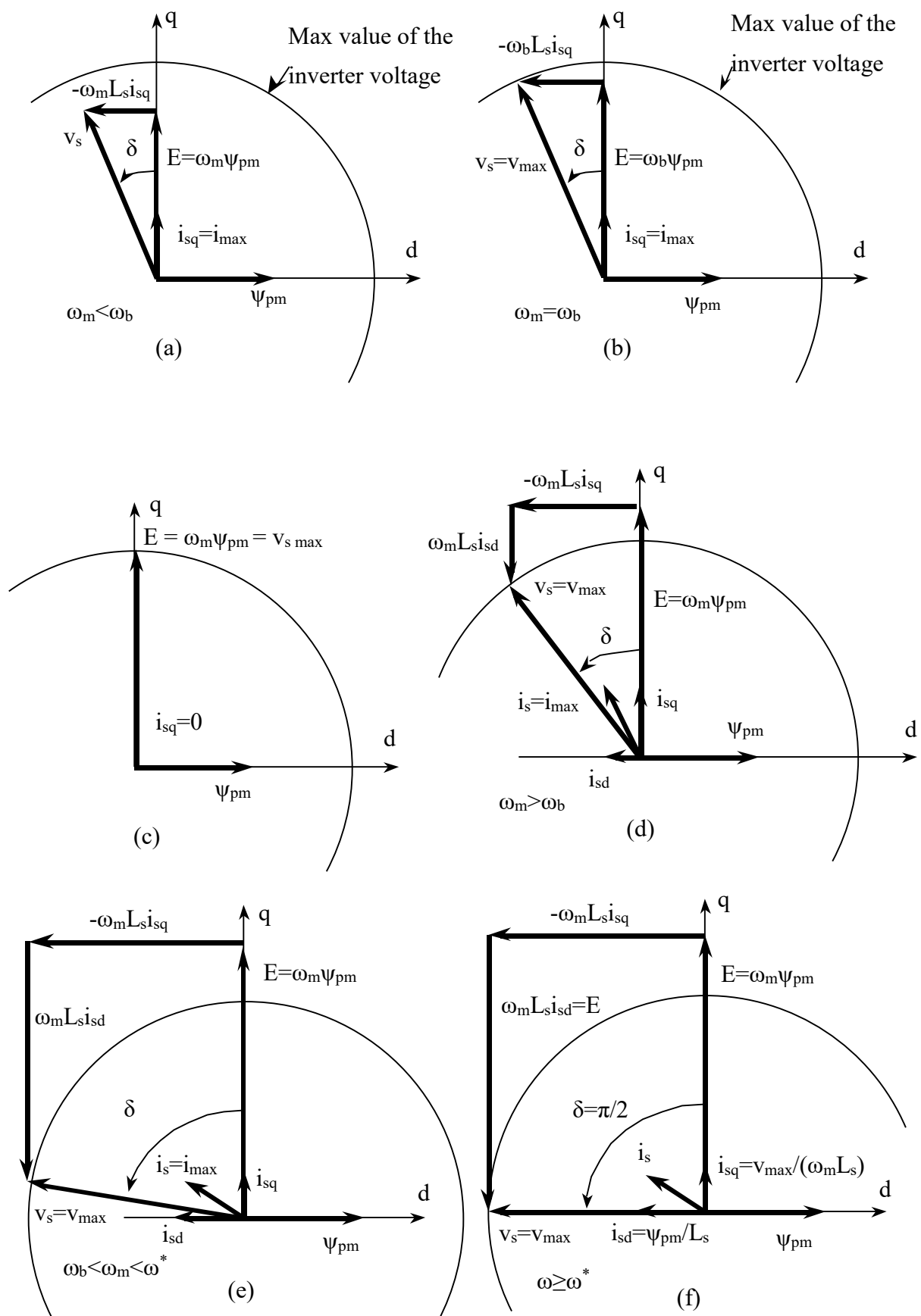


Figure 6-11: Vector diagram as a function of i_{sd} values

You can see (Figure 6-11 (c)) that, in the extreme case of $E=v_{\max}$, if $i_{sd} = 0$, i_{sq} should be zero: this condition is characterized by a speed:

$$\omega' = \frac{E'}{\psi_{pm}} = \frac{v_{\max}}{\psi_{pm}}$$

with zero current and zero torque (no-load speed with $i_{sd}=0$).

On the contrary, with $i_{sd} < 0$, we can get the situation shown in Figure 6-11 (d) where the machine is still able to develop a torque. In this condition, a kind of field weakening is realized (the component i_{sd} is opposite to the flux of the permanent magnets ψ_{pm}). The torque decreases as the value of i_{sq} must decrease to respect the limits on the total magnitude of the stator current $i_{sq} = \sqrt{i_{\max}^2 - i_{sd}^2}$.

Suitably sized motor (using permanent magnets with high coercive force and knee in the third quadrant, with an oversized air gap or internal magnets), you can request a current i_{sd} equal to the ratio between ψ_{pm} and L_s (Figure 6-11 (e)) . The changing point is defined by the value of mechanical speed ω^* . For $\omega > \omega^*$ the voltage $\omega L_s i_{sd}$ is equal to E regardless of mechanical speed. From the Figure 6-11 (f) and looking at the same amplitude of both horizontal axes, the current i_{sq} is equal to $v_{max}/(\omega_m L_s)$ and then the torque is inversely proportional to the mechanical speed.

From this point on, the module of the stator current cannot be maintained at its maximum value but must decrease: $i_s = \sqrt{i_{sd}^2 + i_{sq}^2} = \sqrt{(\psi_{pm}/L_s)^2 + (v_{\max}/\omega_m L_s)^2} < i_{\max}$.

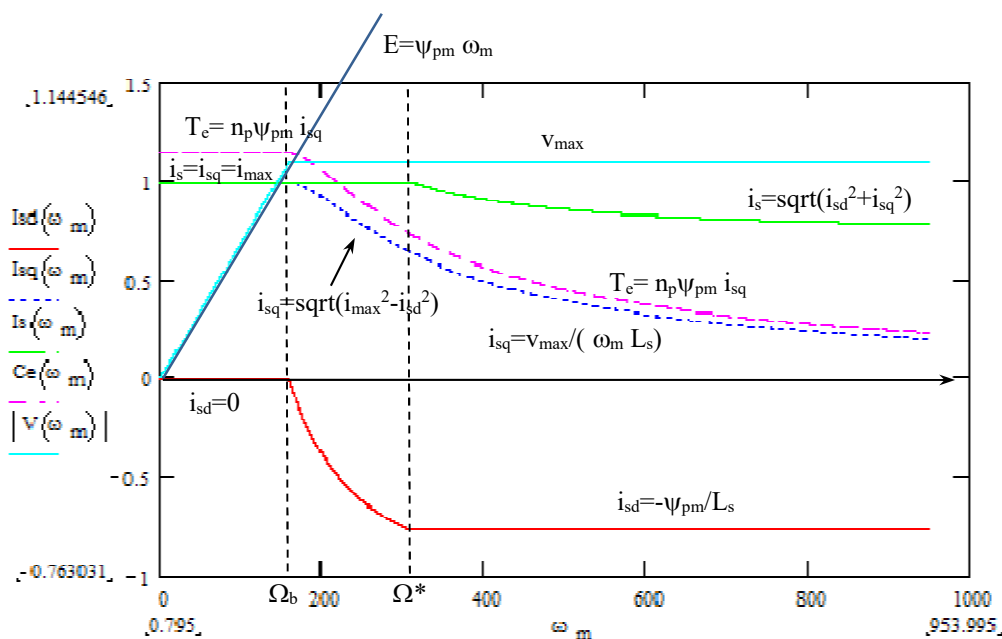


Figure 6-12: AC brushless operating regions

In practice, there are two main areas of application: the most classic of an axis control and the newer type for spindle applications. The first case does not require a speed much higher than the base speed and, therefore, the constant power region is very limited. It is a different story regarding the applications such as spindle. In these cases, the maximum speed can reach considerable values (about 6-8 times the base speed), making these drives a valid alternative to the drives based on induction machines (maximum speed of 5-6 times the base speed). It should be noted that, over the base speed, the emf E continues to grow. At 6 times the base speed E is 6 times the rated E (approximately 6 times the maximum voltage of the inverter). If, in these conditions, the control fails and the current i_{sd} falls to zero, the voltage generated by the machine would destroy static

switches, electrolytic capacitors and the winding insulation. It must provide the system with appropriate protections.

6.6 Control scheme

As seen in the previous section, the controller has to drive the power converter so that it always results $i_{sd} = i_{sd \text{ ref}}$ (depending on the operating region or on a possible voltage controller), while the component i_{sq} must be such that the produced torque is able to maintain the required speed of the machine. The overall scheme of the drive with speed control loop is the following:

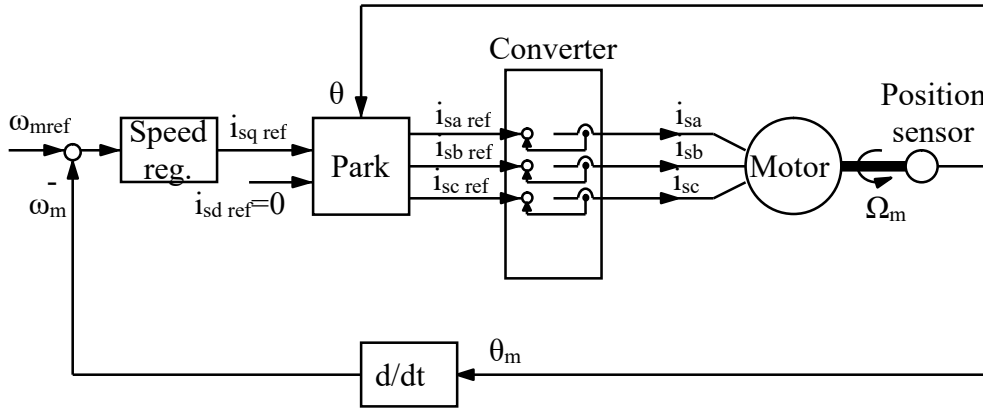


Figure 6-13: Speed control loop (current controlled power converter)

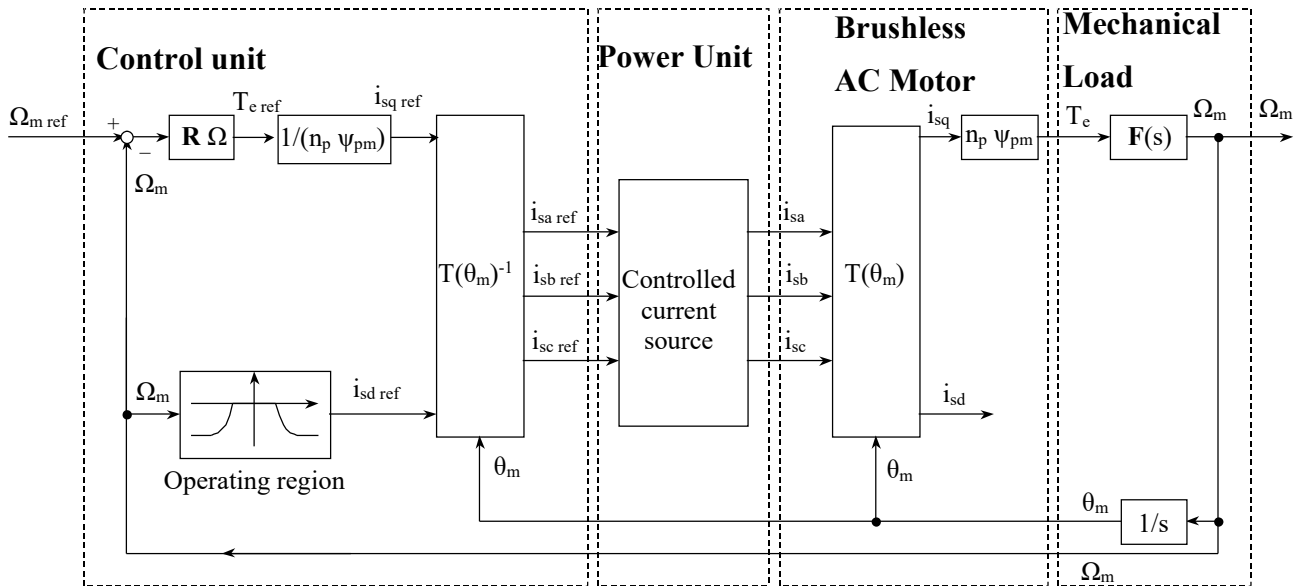


Figure 6-14: Full control scheme (current controller power converter)

In the Figure 6-14, the block $T(\theta_m)^{-1}$ performs the transformation from space phasor variables and phase quantities.

Any position control can be obtained by inserting an appropriate control loop outside of the speed control loop.

Figure 6-14 shows a power supply device with a fast current control that allows the motor currents to follow closely the reference values. Taking into account the stator dynamics (most frequent case), the control scheme changes, as shown in Figure 6-15. In this case the power converter has to establish a voltage reference, while the current regulation is now performed by the controller.

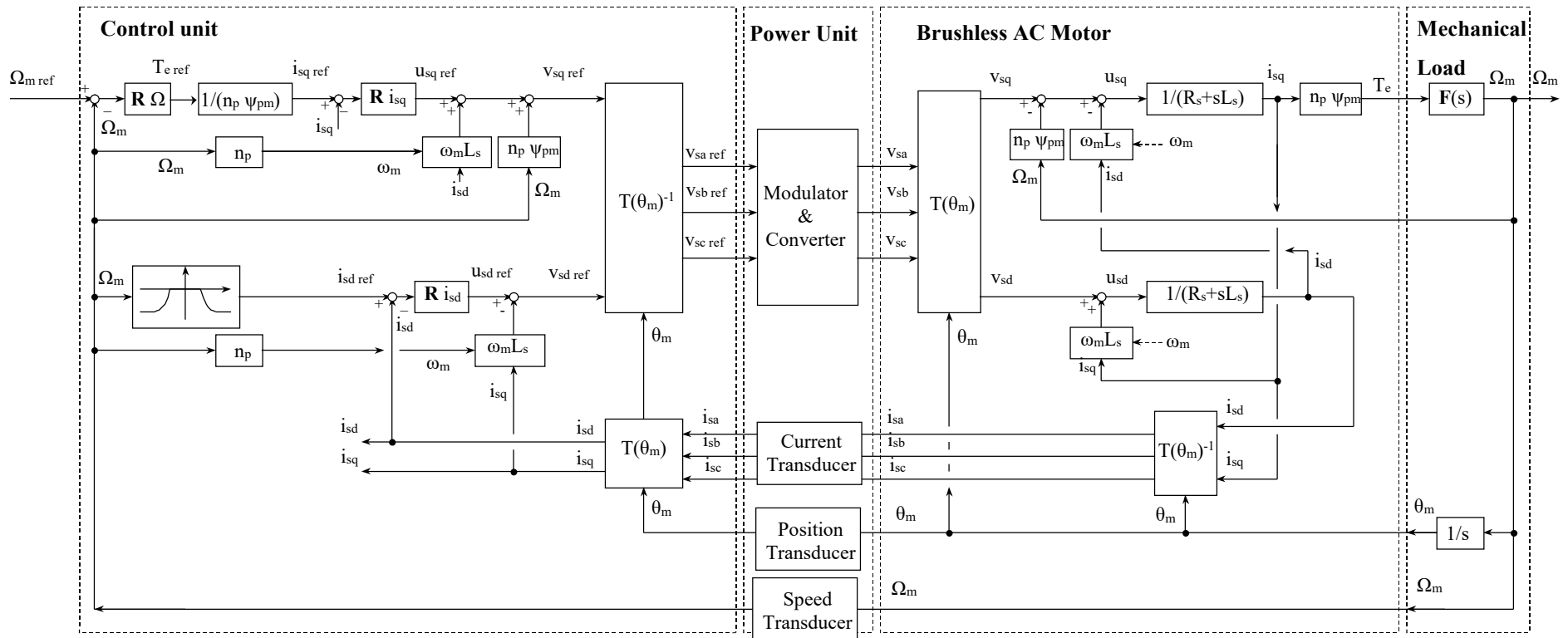


Figure 6-15: Full control scheme (voltage controlled power converter)

The structure of the current controller is based on the differential equations of the machine:

$$\begin{aligned} v_{sd} &= R_s i_{sd} + L_s p i_{sd} - \omega_m L_s i_{sq} \\ v_{sq} &= R_s i_{sq} + L_s p i_{sq} + \omega_m \psi_{pm} + \omega_m L_s i_{sd} \end{aligned}$$

You notice immediately that, in terms of voltages on the two axes, there are different terms: $R_s i_{sd} + L_s p i_{sd}$ (indicated as u_{sd}) and $R_s i_{sq} + L_s p i_{sq}$ (indicated as u_{sq}) are the voltages that actually act on their own current; the terms $-\omega_m L_s i_{sq}$ and $\omega_m L_s i_{sd}$ show the existence of a coupling between the two loops; the term $\omega_m \psi_{pm}$ represents an electromotive force, proportional to the mechanical speed (it has full correspondence with the motional term in the dynamical equations of the armature current of a DC machine). So it is a good thing that the outputs of the current regulators are the reference values of the voltages u_{sd} and u_{sq} . To obtain the reference value of stator voltages, simply add to $u_{s \text{ ref}}$ the corresponding coupling terms and the motional term $\omega_m \psi_{pm}$.

6.7 Current controllers design

With all of the terms of compensation and, if it is reasonable to consider power converter and transducers as pure unity gain, the current regulators can easily be designed on the transfer function: $R_s + sL_s$. ($R_s + sL_d$ e $R_s + sL_q$ in the case of an anisotropic machine).

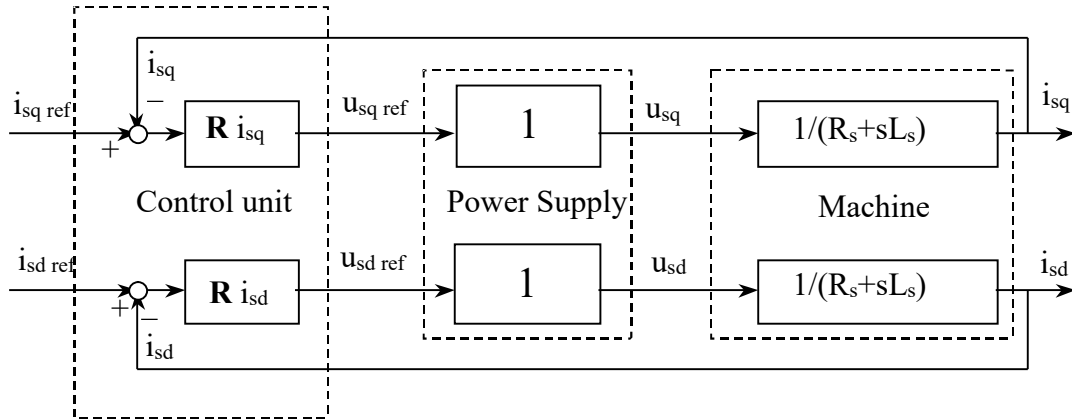


Figure 6-16: scheme for the design of the current controllers

If you do not compensate the coupling terms ($-\omega_m L_s i_{sq}$ and $\omega_m L_s i_{sd}$), the loop controller should "work" even when the reference of the other loop varies. The motional term ($\omega_m \psi_{pm}$), however, would require full compensation through the integral action of the regulator itself; in the case of a machine starting from a speed different by zero, it would lead to some undesirable effects, such as, for example, a sudden break of the machine in the first instant of the operation.

For a better representation of the power supply, the unity gain could be replaced by a delay whose value is tied to the switching time of the converter switches. If the modulator is based on a PWM technique, for example, the delay between the instant of the reference variation and its implementation varies from 0 to the entire period of switching. On average, we can assume a delay equal to one half of the switching time.

6.8 Speed controller design

In order to design the speed regulator it is necessary to know the transfer function of the overall system, described before. The output of the speed controller is the desired value of the electromagnetic torque, which is proportional to the quadrature component of the stator current. The scheme is shown in Figure 6-17.

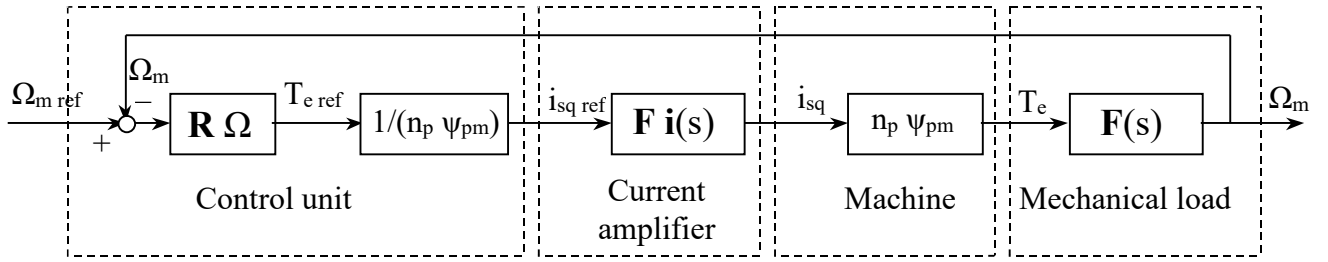


Figure 6-17: scheme for the design of the speed regulator

Again, you can follow two paths. If the bandwidth of the current regulator is very high, compared to that required to the speed control, it can be considered as an ideal current amplifier (unity gain). Otherwise, once designed the current regulator, the transfer function $F i(s)$ is known. The speed controller must therefore be designed considering a system characterized by a transfer function equal to the product of $F i(s)$ for the transfer function of mechanical load $F(s)$.

6.9 Synchronous reluctance machine (SYRM)

Figure 6-18 shows a synchronous reluctance machine with two poles (in Figure 6-20 a four poles machines is presented). It has a typical anisotropic structure of the rotor.

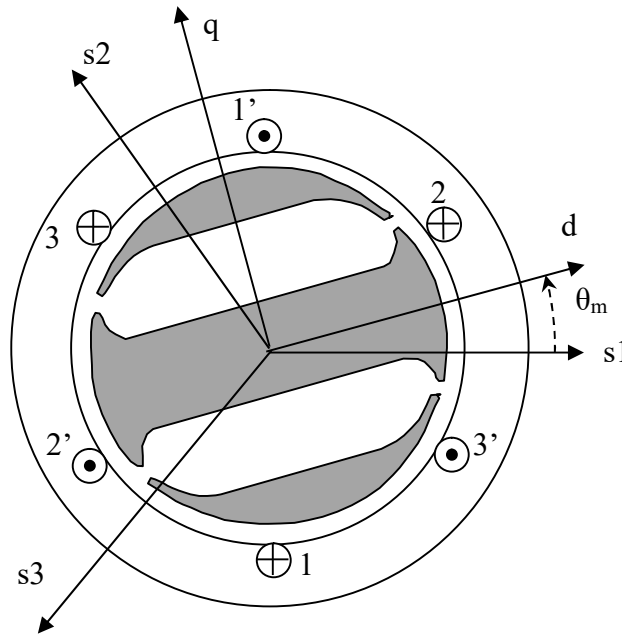


Figure 6-18: synchronous reluctance machine (two poles) $L_d \gg L_q$

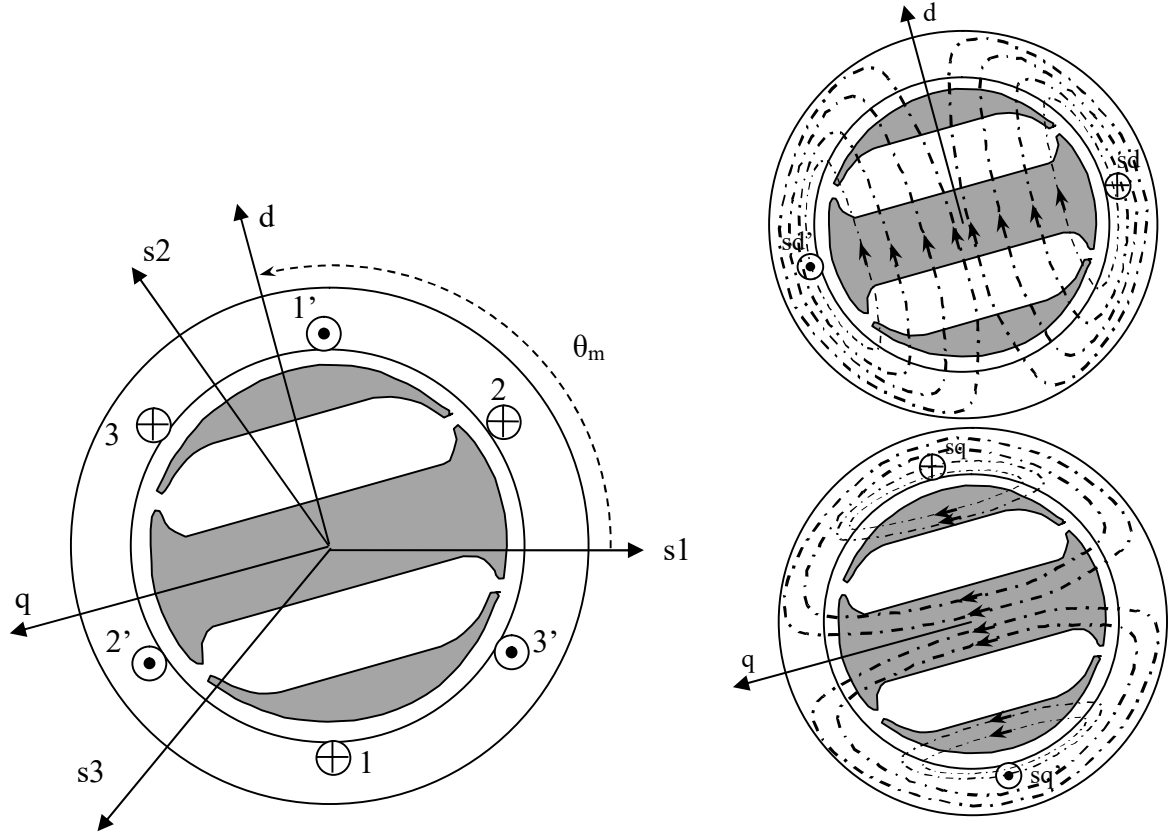


Figure 6-19: synchronous reluctance machine (two poles) with a different choice of the axes: $L_q \gg L_d$; the flux lines due to the current i_{sd} has to pass through large air gaps.

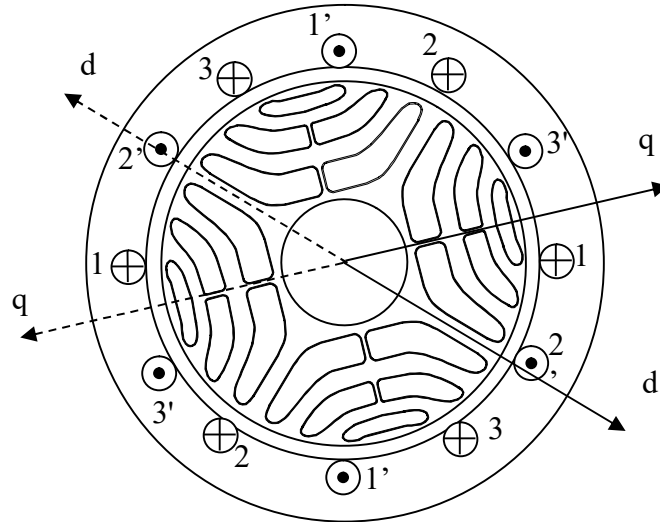


Figure 6-20: Example of a synchronous reluctance machine (four poles)

The stator is equal to the AC brushless stator but the rotor has no permanent magnets and it is designed in order to have a very high difference between the two inductances L_d and L_q ; the higher depends on the choice of the reference frame: in Figure 6-18 $L_d \gg L_q$ while in Figure 6-19 $L_q \gg L_d$. Henceforth the reference as in Figure 6-18 will be adopted.

The torque is only given by its reluctance term:

$$T_e = n_p \left[(L_d - L_q) i_{sd} i_{sq} \right]$$

So it is necessary to feed the machine with both the i_{sd} and i_{sq} currents.

The electrical equations become:

$$v_{sd} = R_s i_{sd} + L_d p i_{sd} - \dot{\theta}_m L_q i_{sq}$$

$$v_{sq} = R_s i_{sq} + L_q p i_{sq} + \dot{\theta}_m L_d i_{sd}$$

There are different control strategies. One among the other comes from the fact that the time constant of the electric circuit on the "d" axis is higher than on the "q" axis ($L_d \gg L_q$ implies $\tau_d \gg \tau_q$).

So a constant value of i_{sd} (equal to the rated value i_{sdn}) is performed, while i_{sq} is used to control the torque. Above the base speed (due to the power converter), it is necessary to decrease i_{sd} .

The control scheme is very similar to the scheme of an AC brushless, shown in Figure 6-15.